

VISVESVARAYA TECHNOLOGICAL UNIVERSITY

“Jnana Sangama”, Belagavi, Karnataka, INDIA



A Project Report
on

“Next-Generation Waste Management: A Smart, Sensor-Based and Community-Driven Approach”

Course Code : 22CSEP76

Submitted in partial fulfillment of the requirement for the award of the degree of

**Bachelor of Engineering
in
Computer Science and Engineering**

Submitted By

HARSHITHA N. K.

1GA22CS071

Under the Guidance of

Prof. Shwetha N
Assistant Professor



**SUSTAINABLE
CITIES AND
COMMUNITIES**

SDG_NO: 11



**Department of Computer Science and Engineering
GLOBAL ACADEMY OF TECHNOLOGY
(An Autonomous Institute Affiliated to VTU)**

Rajarajeshwarinagar, Bengaluru - 560 098

2025 – 2026



GLOBAL ACADEMY OF TECHNOLOGY
(*An Autonomous Institute Affiliated to VTU*)
Department of Computer Science and Engineering



CERTIFICATE

Certified that the project work entitled “**Next-Generation Waste Management: A Smart, Sensor-Based and Community-Driven Approach**” with course code: **22CSEP76**, carried out by **Ms. Harshitha N K** bearing USN **1GA22CS071** bonafide student of Global Academy of Technology in partial fulfillment for the award of **Bachelor of Engineering in Computer Science and Engineering from Visvesvaraya Technological University, Belagavi** during the year 2025-26. It is certified that all corrections/suggestions indicated for Internal Assessment have been incorporated in the Report submitted to the Department. The project report has been approved as it satisfies the academic requirements in respect of the Project work prescribed for the said Degree.

Signature of the
Guide

Signature of the
HoD

Signature of the
Principal

EXTERNAL VIVA

Name of the Examiners

Signature with date

1.

2.



GLOBAL ACADEMY OF TECHNOLOGY
(*An Autonomous Institute Affiliated to VTU*)
Department of Computer Science and Engineering



DECLARATION

I, Harshitha N K bearing USN 1GA22CS071 student of Seventh Semester B.E, Department of Computer Science and Engineering, Global Academy of Technology, Rajarajeshwarinagar Bengaluru, declare that the Project Work “**Next-Generation Waste Management: A Smart, Sensor-Based and Community-Driven Approach**” with course code: **22CSEP76** has been carried out by us and submitted in partial fulfillment of the course requirements for the award of degree in **Bachelor of Engineering in Computer Science and Engineering** from **Visvesvaraya Technological University, Belagavi** during the academic year 2025 – 2026.

Harshitha N K 1GA22CS071

Place: Bengaluru

Date:

ABSTRACT

As urban populations have grown rapidly, the management of solid waste has become more challenging aspects such as inefficient collection, poor segregation of waste, and environmental risks like pollution, health hazards, and a waste of resources have emerged. Traditional waste systems are not equipped with real-time monitoring, automation, and citizen engagement, which results in collection schedules that are less reliable, bins that become crowded, and operational costs that rise.

Through the use of the IoT, smart bins with features such as fill-level detection, instant alerts, and cloud-based tracking can be more responsive as well as waste can be managed through data processing. To address this, “Next-Generation Waste Management: A Smart, Sensor-Based and Community-Driven Approach”, the Sortify web-app, is launched. Sortify web-app is an IoT-powered smart waste management platform that unites sensor-enabled bins, cloud-based data storage, and database-linked bin locations to optimize waste collection. In the case when the capacity of a bin is full, the driver who is in charge is sent the pickup notification through the web interface so that the collection is made on time and overflow avoided.

In addition, Sortify web-app has a mechanism for rewarding citizens that encourages them to participate in the handling of waste and in positive environmental behavior. The integration of automation, real-time monitoring, and community engagement makes Sortify web-app a practical and innovative solution for modern cities. It promotes responsible waste management practices and strengthens collaboration between citizens and local authorities for a cleaner urban environment. By using data-driven insights, Sortify web app helps optimize routes and reduce unnecessary trips. It contributes to better resource utilization and transparency in waste management operations. Overall, Sortify web-app represents a step forward in building smarter and more sustainable urban communities.

ACKNOWLEDGEMENT

The satisfaction and the euphoria that accompany the successful completion of any task would be incomplete without the mention of the people who made it possible. The constant guidance of these persons and encouragement provided, crowned our efforts with success and glory. Although it is not possible to thank all the members who helped for the completion of the Mini Project individually, we take this opportunity to express our gratitude to one and all.

I are grateful to the Management and our institute **GLOBAL ACADEMY OF TECHNOLOGY** with its very ideals and inspiration for having provided us with the facilities, which made this, Mini Project a success.

I express our sincere gratitude to **Dr. Balakrishna H. B.**, Principal, Global Academy of Technology, for the support and encouragement.

I wish to place on record, our grateful thanks to **Dr. Kumaraswamy S**, Professor and Head, Department of CSE, Global Academy of Technology, for the constant encouragement provided to us.

I are indebted with a deep sense of gratitude for the constant inspiration, encouragement, timely guidance, and valid suggestions given to us by our guide **Prof. Shwetha N, Assistant Professor**, Department of CSE, Global Academy of Technology.

I are also thankful to the Project Coordinator, **Prof. Aparna K S** and all the staff members of the department for providing relevant information and helping in different capacities in carrying out this Mini Project.

Last, but not least, we owe our debts to our parents, friends and also those who directly or indirectly have helped us to make the Mini Project work a success.

Harshitha N K

1GA22CS071

CO-PO-PSO MAPPING AND JUSTIFICATION OF THE PROJECT

Course: Major Project Phase II	No. of Credits:06
Course Code: CSE23706	CIE Marks: 100, SEE marks:100

Course Outcomes (CO)

Course outcomes	Descriptions
CO 1	Understand the requirements of the end user considering economic, social and environmental factors for providing a feasible solution with strong basics of Computer Science subjects.
CO 2	Design a feasible solution for the identified problem with good literature survey using modern tools.
CO 3	Prepare well-structured report of the project with plagiarism check and communicate the same in different phases / journals/ conferences / project exhibitions.
CO 4	Coordinate the assigned task with the team members within in specified time in concern with the project guide.

Program Outcomes (PO):

PO1: Engineering Knowledge PO2: Problem Analysis PO3: Design/Development of Solutions PO4: Conduct Investigations of Complex Problems PO5: Modern Tool Usage PO6: The Engineer and Society	PO7: Ethics PO8: Individual and Team Work PO9: Communication PO10: Project Management and Finance PO11: Life-long Learning
---	---

CO – PO Mapping:

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	3	3				3	3				1	3	1
CO2			3	2	3						2	3	2
CO3								2			2	2	1
CO4								3	3	2	2	1	1

Program Specific Outcomes (PSO):

PSO1: Design, implement, and test system software and applications to meet real-world requirements.

PSO2: Develop effective solutions in communication networks, database systems, and computing applications

CO- PO-PSO Justification

CO1: *Understand the requirements of the end user considering economic, social and environmental factors for providing a feasible solution with strong basics of Computer Science subjects.*

PO/PSO	Grade	Justification
PO1	3	Applied core engineering and computer science fundamentals to solve a real-world waste management problem.
PO2	3	Analysed community waste issues and defined clear system objectives for effective segregation and monitoring.
PO6	3	Considered public hygiene, safety, and societal needs while designing the smart waste management system.
PO7	3	Supported environmental sustainability through automated waste segregation and clean-city practices.
PO11	1	Demonstrated basic project planning and task execution within the given academic timeline.
PSO1	3	Designed and implemented an IoT-based software and hardware solution to meet real-world user requirements
PSO2	1	Used basic communication and database concepts to support system data flow and monitoring.

CO2: Design a feasible solution for the identified problem with good literature survey using modern tools.

PO/PSO	Grade	Justification
PO3	3	Designed and prototyped a feasible smart waste segregation system meeting real-world requirements.
PO4	2	Conducted systematic testing and validation to improve system accuracy and performance.
PO5	3	Used modern tools such as Arduino, ThingSpeak, Supabase, and web dashboards to implement the solution.
PO11	2	Followed structured project planning and task execution within the defined timeline.
PSO1	3	Built an end-to-end IoT-based software and hardware system for practical deployment.
PSO2	2	Implemented cloud communication and database interactions for real-time monitoring.

CO3: Prepare well-structured report of the project with plagiarism check and communicate the same in different phases / journals/ conferences / project exhibitions.

PO/PSO	Grade	Justification
PO8	2	Documented ethical practices and ensured responsible system development.
PO11	2	Improved documentation quality through continuous learning and feedback.

PSO1	2	Prepared a structured report explaining system design, implementation, and results.
PSO2	1	Included basic database and communication details in project documentation.

CO4: Coordinate and execute the assigned task and evaluate with the team members within in specified time in concern with the project guide.

PO/PSO	Grade	Justification
PO8	3	Worked collaboratively by assigning responsibilities and resolving issues effectively as a team..
PO9	3	Coordinated project execution through clear communication and regular review discussions.
PO10	2	Managed tasks according to the project timeline to meet submission deadlines.
PO11	2	Improved execution through continuous self-learning and guidance from the project mentor.
PSO1	1	Assisted in implementing and reviewing system components during project development.
PSO2	1	Provided basic support in handling communication and database-related tasks.

TABLE OF CONTENTS

Chapter No.	Particulars	Page. No
	Abstract	i
	Acknowledgment	ii
	Table of contents	iii
	List of Figures	v
	List of Tables	vi
	Glossary	vii
1	Introduction	1
	1.1 Definitions	1
	1.2 Project Report Outline	1
2	Review of Literature	3
	2.1 System Study	4
	2.2 Proposed Work	4
	2.3 Scope of the project	5
3	System Requirement Specification	6
	3.1 Functional Requirements	6
	3.2 Non-Functional Requirements	7
	3.3 Hardware Requirements	7
	3.4 Software Requirements	8
4	System Design	10

	4.1 Design Overview	10
	4.2 System Architecture	11
	4.3 Data Flow Diagrams	13
	4.3.1 Data Flow Diagram – Level 0	13
	4.3.2 Data Flow Diagram – Level 1	13
	4.3.3 Data Flow Diagram – Level 2	14
	4.3.4 Data Flow Diagram – Level 3	15
	4.3.5 Data Flow Diagram – Level 4	15
	4.4 Use Case Diagram	16
	4.5 Class Diagram	17
	4.6 Sequence Diagram	18
	4.7 Collaboration Diagram	19
	4.8 Activity Diagram	21
	4.9 Modules	22
	4.9.1 Waste Segregation Module	22
	4.9.2 Cloud Integration Module	22
	4.9.3 Admin Dashboard Module	23
	4.9.4 User Dashboard Module	23
	4.9.5 Driver Dashboard Module	24
5	Implementation	25
	5.1 Objective 1	25
	5.2 Objective 2	25
	5.3 Objective 3	26
	5.4 Results and Discussions	26

6	Testing	27
	6.1 Test Environment	27
	6.2 Unit Testing	28
	6.3 Integration Testing	29
	6.4 System Testing	29
	6.5 Functional Testing	29
7	Results	30
	Conclusion	34
	Bibliography	35
8	Appendix: SDGs, Publication Proceedings-Communication and Certificates	

LIST OF FIGURES

Figure No.	Figure Name	Page. No
Figure 4.2	System Architecture	11
Figure 4.3.1	Data Flow Diagram- level 0	13
Figure 4.3.2	Data Flow Diagram- level 1	13
Figure 4.3.3	Data Flow Diagram- level 2	14
Figure 4.3.4	Data Flow Diagram- level 3	15
Figure 4.3.5	Data Flow Diagram- level 4	15
Figure 4.4	Use Case Diagram	16
Figure 4.5	Class Diagram	17
Figure 4.6	Sequence Diagram	18

Figure 4.7	Collaboration Diagram	19
Figure 4.8	Activity Diagram	21
Figure 7.1	Waste Segregation System – Hardware Prototype	30
Figure 7.2	Metal Waste Detection Using Proximity Sensor	30
Figure 7.3	Wet Waste Detection Using Rain Drop Sensor	31
Figure 7.4	Dry Waste Detection When Metal & Wet Sensors Are False	31
Figure 7.5	Final Output–Segregated Waste in Individual Bins	31
Figure 7.6	Sortify Web Application – Landing Page and Key Features	32
Figure 7.7	User Dashboard of the Sortify Web Application	32
Figure 7.8	Admin Dashboard of the Sortify Web Application	32
Figure 7.9	Driver Dashboard With GSM Alert Integration	33

LIST OF TABLES

Table No.	Table Name	Page. No
Table 6.2	Unit test cases	28

CHAPTER 1

INTRODUCTION

1.1 Definitions:

- **Smart Dustbin:** Sensor based smart dustbin which does waste segregation and monitors the fill level of sub-bin, data is sent to ThingSpeak & drivers are notified if it is full.
- **Waste Segregation:** Automatic sorting of waste based on wet, dry, metallic and recyclable materials based on sensor such as camera/object detection, proximity switch, metallic sensor, raindrop sensor.
- **Ultrasonic Sensor:** Measuring bin fill level and triggers alerts. Through the GSM module data is sent to ThingSpeak.
- **ThingSpeak IoT Cloud:** ThingSpeak is a platform to receive, store and visualize the Real-time data of the sensors from the smart dustbin.
- **Supabase Database:** Backend database for the Sortify web app used for storing users, bins, truck assignments, reports and WebRTC communication data.
- **Admin Dashboard:** Panel for authorities to monitor bin statuses from ThingSpeak, assign drivers, and manage collection activities using Supabase data.
- **Driver Dashboard:** Interface for the driver to access the assigned bins, navigate, get real-time alerts and update the collection status.
- **User Dashboard:** Portal for residents meant to report issues, track trucks and check the bin status using Supabase with backing data.
- **IR Sensor (Infrared Sensor):** IR Sensor is used to find out the presence of an object and it helps in identifying the waste in the waste segregation.
- **Segregation Mechanism:** System in which servo motor (flaps) and stepper motor (bin rotation) are implemented for controlling the waste for the correct category.
- **Communication Module (GSM):** Transmits sensor data to ThingSpeak and also gives SMS alerts to drivers when bins are full.
- **Cloud Integration:** Cloud integration is being done using ThingSpeak for IoT data and Supabase for application/database operations within the Sortify system.

1.2 Project Report Outline

- **Chapter 1: Introduction**

Introduces the project topic, background, problem statement, objectives, and relevance. It also includes key definitions and a summary of the report structure.

- **Chapter 2: Literature Survey**
Discusses extant research, technologies, and systems stakeholders in the project area. Identifies the gaps in existing solutions and establishes the need for the proposed system.
- **Chapter 3: System Requirement Specification**
Specifies functional requirements and non-functional requirements of the system. It also includes the required hardware and software required for development and deployment.
- **Chapter 4: System Design**
Describes the overall system architecture and designs the modules and various diagrams such as data flow, use case, and class diagrams. It visualizes the interaction between the components in the system.
- **Chapter 5: Implementation**
Describes implementation of IoT segregation module, ThingSpeak cloud integration, Supabase-based dashboard, SMS alert mechanism, role-wise web functionalities, i.e. Admin, User, Driver.
- **Chapter 6: Testing**
Covers the testing strategies used including test environment setup, unit testing, integration testing, system testing and functional testing to verify the correctness of the all the components.
- **Chapter 7: Results**
Presents the results of the implemented system, in which the functions of segregation accuracy, SMS alerts, and dashboard functions are presented in an evaluation.
- **Conclusion**
Summarizes the project's accomplishments, and suggests ways to make improvements.
- **Bibliography**
List the references, research papers, articles, websites and tools referred to in the course of the project.

CHAPTER 2

LITERATURE SURVEY

In recent years, IoT data-driven services have been an active area of research and development by researchers and developers.

The paper "Innovative Waste Management: The Efficiency of IoT-Enabled Smart Bins" (2024) by Shiva Mehta et al. presents a smart bin management system integrated with ultrasonic sensors, weight sensors, temperature sensors, and tilt sensors using a strong software model to process information in real time and optimize the route. The system reported to have a 50% improvement in collection efficiency with a reduction in bin overflow by 83.3%, significantly improving the public satisfaction of the services and decreased operational costs

The research "IoT-Based Smart Waste Management System with Level Indicators for Effective Garbage Waste Segregation" studying in 2024 by Mary P. Varghese et al. leads the implementation of smart waste management system using sensor-equipped waste management bins and long-range radio (LoRa) communication modules to ensure proper fill-level detection activities, route planning, and waste management. The system promotes participation in recycling and mitigates the risk of overflow, helping to fight urban sustainability measures as well.

In "Garbage Monitoring System using IoT" (2024), Mithun P. et al. propose a cloud integrated waste monitoring system using ultrasonic & IR sensors and thus send real-time data in MQTT/CoAP protocol optimizing collection routes and mobile-based alerts for authorities. The system also cuts down on Green House Gas emissions by keeping redundant collection trips to a minimum.

The research work "IoT-Based Smart System for Garbage Detection and Segregation" (2023) by Dr. M.V. Rajesh et al. proposes an intelligent dustbin solution with various sensors such as ultrasonic, moisture, and proximity sensors and automated waste segregation and robotics solution for navigation to dumping zones. The solution encourages recycling, saves on labor and helps in the closed-loop waste management.

Lastly, "IoT-Enabled Smart Waste Management: Leveraging the Power of IoT for Sustainable Solutions" (2023) by Righa Tandon et al. is a multi-layered architecture where IoT(binsensors), Fog(collector/culator edge-processing) and Cloud(public cloud-enabled processing and decision-making) layers serve as the basis for scalable predictive waste analytics with alerting mechanism. This waste management system brings in combination artificial intelligence, machine learning, real-time sensor data to automate the waste collection process, improve route planning, and contribute to sustainable city infrastructure.

2.1 System Study

The latest progress in intelligent waste management technologies has been on automating bin tracking, enhancing the accuracy of the segregation process, and cloud-based tracking. Contemporary IoT-powered applications operate on the basis of sensors, which can be ultrasonic, IR, or moisture sensors attached to microcontrollers to assess the level of fill and sort waste. Such systems normally post data to cloud systems via the IoT protocols and dashboards can be used to show real-time bin conditions.

Nevertheless, most of the systems that have been developed are independent systems that either concentrate on monitoring fill-level only or on some basic waste detection without incorporating user involvement and real-time dashboarding and alarms. There are solutions which offer segregation but are not connected to the cloud and ones which track the status of the bins but leave waste separation to be performed by hand.

Community-facing platforms are not implemented extensively where citizens can engage with the waste system and either comment or see the insights on the environment. Likewise in most systems, driver-side communication may be limited or not available, and must frequently be updated manually.

Sortify fills these holes with automated segregation, cloud-monitored with ThingSpeak, user interaction via a custom web dashboard, driver notifications via GSM-based SMS, and administration via Supabase. This forms a single system in which waste monitoring and cloud visualization: both systems interact with humans.

2.2 Proposed work

The proposed project title "**Next-Generation Waste Management: A Smart, Sensor- Based and Community-Driven Approach**" introduces **Sortify**, a smart waste management and segregation system that will be web-based. It adds an integrated platform, which automates the process of segregating waste and also allows users to monitor it in real time and have a say in it.

At the heart of the system is a sensor-equipped smart dustbin that has separate compartments for the **wet, dry and metallic waste**. Sensors, such as ultrasonic sensors, infrared sensors, raindrop sensors and metallic proximity sensors, identify type of waste and the status of bins at a level. The micro controller in turn processes these readings and segregates mechanically by servo and stepper motors.

All the sensor values and bin-level data are sent to the **thing speak IoT cloud** that is meant to save and visualize the data by real-time gauges and graphs. When the bin is full in capacity the GSM module automatically sends an SMS alert to the driver.

A three-role web application complements the IoT system:

- **Admin Dashboard:** Allows administrators to monitor bin levels through embedded ThingSpeak visualizations and manage driver accounts via Supabase.
- **User Dashboard:** To facilitate citizen to register, submit feedback, appreciate segregation trends with ThingSpeak Graph, interact with reward checklist.
- **Driver Dashboard:** Provides drivers with bin-level gauges from ThingSpeak and SMS alerts for full-bin notifications.

The proposed system in this manner combines physical segregation, cloud visualization and human centered dashboards into one unified workflow.

2.3 Scope of project

The extent of Sortify is to create a full IoT to cloud waste management ecosystem which will automate the process of segregation, facilitate real-time monitoring and foster citizen participation. The solution is suited for the urban environment where accuracy of waste separation and timely collection are key factors to ensure hygiene and sustainability.

The system supports:

- Automated wet/dry/metallic segregation using sensors
- Cloud and locator: "Software that observes in real time the status of a cloud and manages the information and delivers alerts to ThingSpeak."
- SMS-based notifications to drivers using a GSM module
- Web-based dashboards for admin, user, and driver access
- User participation through feedback and reward mechanisms
- Centralized management through Supabase

Because the architecture is modular and integrated to the cloud, it can scale easily - meaning more bins can be accommodated, more users and more dashboards can be added without making significant changes. The system removes the need to physically monitor the system and allows for rapid responses via cloud visualisation and immediate driver alerts.

Sortify makes cities cleaner and enables responsible waste habits and has created a baseline for further expansions, including predictive analytics, smart dashboards and smarter decisions around waste.

CHAPTER 3

SYSTEM REQUIREMENT SPECIFICATION

3.1 Functional Requirements

1. Waste Detection and Segregation

- The system must detect waste using IR, raindrop, and metallic sensors.
- The system must classify waste into wet, dry, or metallic categories.
- The segregation mechanism must route waste to the correct compartment.

2. Bin Level Monitoring

- The ultrasonic sensor must measure the bin fill level in real time.
- The microcontroller must send sensor values to the ThingSpeak cloud.

3. Cloud Communication

- The system must upload waste type and bin-level data to ThingSpeak.
- Dashboards must fetch graphs and gauge levels from ThingSpeak.

4. Driver Notification

- When the bin is full, the GSM module must send an SMS alert to the driver.

5. User Dashboard Functions

- Users must be able to register and log in via Supabase.
- Users must view segregation graphs (ThingSpeak).
- Users must view an interactive bin map.
- Users must submit feedback, which is stored in Supabase.
- Users must interact with the reward checklist.

6. Admin Dashboard Functions

- Admin must log in via Supabase.
- Admin must view bin-level gauge charts (ThingSpeak).
- Admin must create driver accounts and auto-generate passwords.
- Admin must view user feedback from Supabase.

7. Driver Dashboard Functions

- Drivers must log in via Supabase.
- Drivers must view bin-level gauges from ThingSpeak.
- Drivers must receive SMS alerts for full bins.

3.2 Non-Functional Requirements

1. Performance Requirements

- Sensor readings must update on ThingSpeak with minimal delay.
- Dashboards must load graphs and gauges within acceptable response time.

2. Reliability

- The IoT unit must operate continuously and provide consistent data.
- GSM alerts must always trigger upon full-bin detection.

3. Usability

- Dashboards must have a simple, intuitive interface.
- Users should be able to navigate feedback, map, and reward features easily.

4. Scalability

- The system must support adding new bins, drivers, and users without major changes.
- ThingSpeak and Supabase should handle additional data points smoothly.

5. Security

- Authentication must be handled securely using Supabase Auth.
- Only authorized admins should create driver accounts or view system feedback.

6. Availability

- Cloud dashboards must remain accessible 24/7.
- ThingSpeak APIs must be reachable for real-time updates.

7. Maintainability

- The system must allow easy updates to code, sensors, or dashboards.
- Modular architecture should simplify debugging and expansion.

3.3 Hardware Requirements

Hardware requirements outlines the physical equipment required to either construct and to run the system effectively. It has the features of specifications for embedded components in smart bins, system hardware for dashboard control, and real-time monitoring.

A. Embedded Smart Bin Hardware

- **Microcontroller:**
 - Arduino Uno: Used for reading sensor data, processing waste classification logic, and controlling motors involved in segregation
- **Sensors and Modules:**
 - IR Sensor: Used to detect the presence of incoming waste at the bin opening.
 - Proximity Switch: Detects presence of metal.
 - Raindrop Sensor: Detects moisture in waste.

- Ultrasonic Sensor: Measures bin fill level.
- **Actuators and Mechanisms:**
 - Stepper Motor: Used to control the internal mechanism of sorting better and means for waste.
 - Servo Motor: Will control the bin lid in both the open and close direction.
 - Shaft Adapter: For connecting the motors to the rotating components for performing mechanical operations.
- **Power and Wiring:**
 - Batteries: To power the phone for mobile or remote use.
 - Jumper Wires: Jumper Wires are used to connect the circuits across the components.
- **Communication Module**
 - GSM Module: On full bin Sends alert to the driver via SMS automatically.

B. System Hardware for Web App and Cloud Dashboard

- **Processor (CPU):**
 - Minimum: Intel Core i3 or equivalent
 - Recommended: Intel Core i5 / AMD Ryzen 5 or higher for multi-dashboard operations / Apple M1 or later (e.g., M2, M3)
- **RAM:**
 - Minimum: 4 GB
 - Recommended: 8 GB or higher for development tasks
- **Storage:**
 - At least 100–500 MB required.
 - SSD recommended for faster loading and development speed
- **Internet Connection:**
 - Stable Wi-Fi or 4G/5G network required for IoT communication and live dashboard updates.

3.4 Software Requirements

Software requirements refer to requirements in programs, tools, and platforms required for system development and operation. This involves programming languages, frameworks, libraries, and supporting environments that are needed to program both the IoT-based waste segregation system and the Sortify web application.

A. Frontend Development

- **HTML5:** Structure and layout of all dashboard pages.
- **CSS3 /Custom Styling:** Styling for UI components and responsive layouts.
- **JavaScript:** Handles dynamic interactions, API calls to ThingSpeak and Supabase.

- **Embedded ThingSpeak Widgets:** Displays bin-level gauges and segregation graphs in dashboards.
- **Leaflet.js (Open-source map library):** Used to display the interactive map in the User Dashboard using **OpenStreetMap tiles**

B. Backend & Database

- **Node.js (JavaScript Runtime Environment):** Used for backend logic, API routing, handling requests from dashboards, and server-side operations.
- **npm (Node Package Manager):** Required for installing backend dependencies and running the application (e.g., npm install, npm run dev).
- **Supabase:** Utilized as the core backend platform, offering essential services such as:
 - **Authentication** (Admin, User, and Driver login & registration(only for user))
 - **Database for storing users, driver accounts, feedback, reward data.**
 - **API endpoints for CRUD operations.**
- **ThingSpeak IoT Cloud**
 - Stores real-time sensor values (waste type + bin-level data)
 - Provides graphs and gauge visualizations for dashboards
- **Microcontroller Firmware (Arduino IDE):**
 - Pushes sensor data to ThingSpeak.
 - Communicates with GSM module for SMS alerts

C. Development Tools

- **Visual Studio Code (VS Code):** Main code editor for frontend, backend, and microcontroller development.
- **Browser Developer Tools (Chrome/Edge):** Debugging frontend UI and JavaScript functionalities.
- **Arduino IDE:** Programming the microcontroller and testing sensor code

D. Operating System

- **OS Compatibility:**
 - Windows 10 / 11, macOS, or Linux distributions.

CHAPTER 4

SYSTEM DESIGN

4.1 Design Overview

The design of the Sortify web application is focused on developing a practical, automated, and community supporting waste management system. It uses an IoT-enabled smart segregation bin, monitor, and role-specific dashboard on the cloud that allows real-time data updates, efficient communication, and active citizens.

The system combines the sensor-driven hardware for class bin-level waste classification and a web platform for the visualization of bin status, driver management and submission for feedback and the promotion of responsible waste behavior. The main concept is ensuring continuous smooth data flow between the physical smart bin and the dashboards, accuracy, responsiveness to be accurate, and modular operation.

The architecture consists of the following main components:

- **Smart Bin Module:**

This module comprises a complete IoT hardware setup that is responsible for detecting and segregating waste. The bin has sensors like IR, ultrasonic, raindrop and metallic proximity sensors which detect the type of waste and track the fill levels. A microcontroller manages the servo and stepper motors to sort the waste into the corresponding compartment (wet, dry or metallic). All sensor readings and bin-level information is constantly processed in preparation to send to the cloud. Upon the bin's full capacity, the system sends an alert message through the GSM module to a user through an SMS.

- **Cloud Communication Layer:**

This layer is what makes for one hundred and to a hundred percent communication between the IoT bin and the web dashboards initiate sure. All sensor values are sent to the ThingSpeak IoT cloud that stores and visualizes real-time data in the form of charts and gauges. Supabase is used as the backend database for the application where authentication, user accounts, driver records, feedback submissions, and reward checklist mechanism are handled in the app. Together, ThingSpeak and Supabase constitute the backbone of the data storage, building visualization and the application level communication.

- **Admin Dashboard Module:**

The Admin Dashboard is the monitoring and management center. It incorporates ThingSpeak widgets to impose real-time bin-level gauges and segregation trends. Administrators can set up driver accounts via Supabase and be able to view feedback submitted by users. This

module enables authorities to maintain a record of the condition in bins and manage the operation of the system efficiently.

- **Driver Dashboard Module:**

The Driver Dashboard creates access to basic and essential interface for field staff. The drivers can login through Supabase and view the live bin-level gauges from ThingSpeak. Additionally, drivers receive SMS alerts via the GSM module in case the bin becomes full and can be alerted immediately without having to rely on complex navigation or route-based features.

- **User Dashboard Module:**

The User Dashboard boosts active participation of community members. Users can sign up or log in with Supabase authentication, access segregation graphs that were taken from ThingSpeak, and access an interactive map to see the location of bins. They can also submit feedback and communicate with a reward-based checklist system that can promote responsible waste habits. This module enables the community to become well-informed and to make contributions to a cleaner environment.

The modular nature of the system permits each of the component IoT hardware, cloud communication and web dashboards to be developed, tested and maintained independently. By combining automated segregation, real-time monitoring, and user engagement, the Sortify web app is part of an efficient and scalable solution that is community-centered and appropriate for the modern city waste management system.

4.2 System Architecture

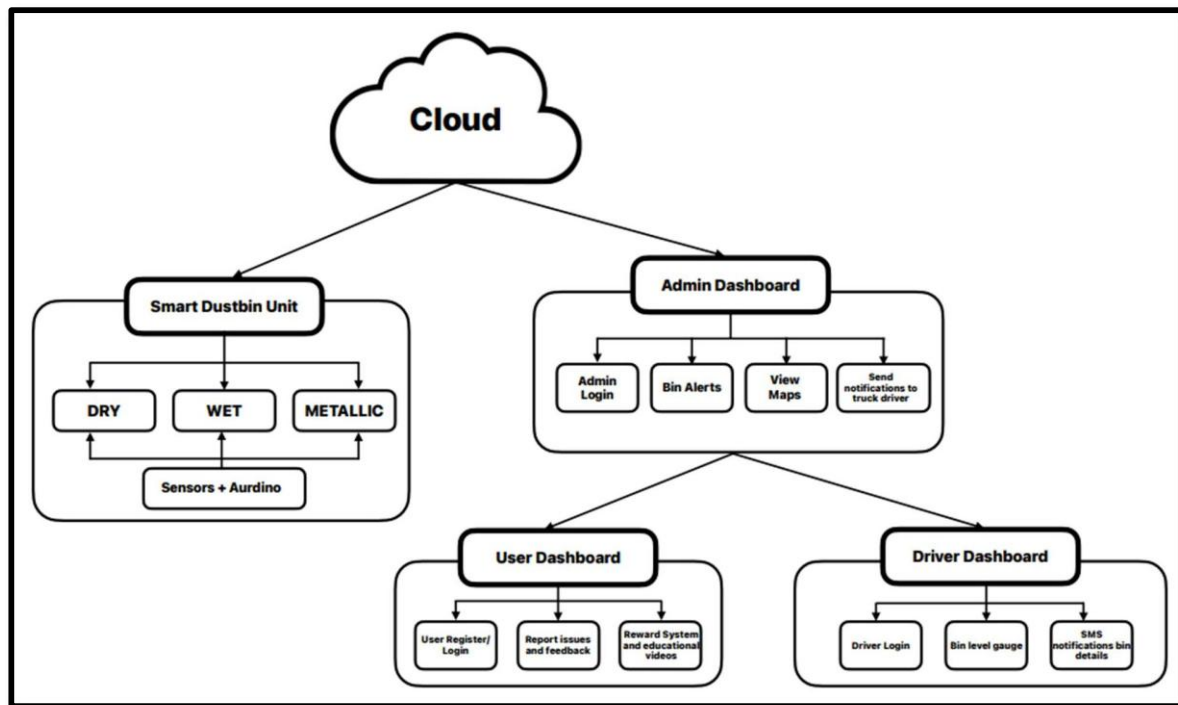


Fig 4.2 System Architecture

Sortify platform system architecture plays the role of capturing and storing waste using smart dustbin IoT devices coupled with cloud services and role-based web dashboards. The architecture is divided into three major layers, namely IoT hardware layer, cloud layer, and application/dashboard layer.

At the physical layer **Smart Dustbin Unit** has separate mechanisms of wet, dry and metallic waste. Waste detection and classification is carried out with help of various sensors like ultrasonic sensor (bin level), IR sensor (object detection), raindrop sensor (wet waste) and metallic proximity sensor. These sensors connect with a microcontroller which does the reading processing and controls the segregation mechanism.

The microcontroller sends data in real-time to **ThingSpeak** (acting as the IoT cloud layer) via bin level and waste-type data. ThingSpeak has the capability of storing and visualizing sensor data which gives gauge indicators and charts for all dashboards to fetch. In parallel, when the bin is full, the microcontroller sends an alert to the **GSM module** to send an SMS alert directly to the driver having the bin location information.

On the application end, the **Admin Dashboard** fetches live data of the bin-level gauges from ThingSpeak and presents this data in a form of embedded visualizations. It also interacts with **Supabase**, which is used to manage the account of the drivers and get user feedback. Admins have access to creating new driver, see details of drivers, and all feedback submitted by users.

The **User Dashboard** is used for user registration and login using Supabase and access to an interactive map with pinpoints of bin location. Users are able to see segregation graphs (fetched from ThingSpeak), report problems/feedback, and interact with a reward checklist system that is managed by Supabase.

The **Driver Dashboard** has been developed with minimal functionality. Drivers log-in using Supabase and see the current bin level gauges fetched from ThingSpeak. At the same time, they get SMS messages from the GSM module every time a bin is full.

4.3 Data Flow Diagrams

4.3.1 Data Flow Diagram- LEVEL 0

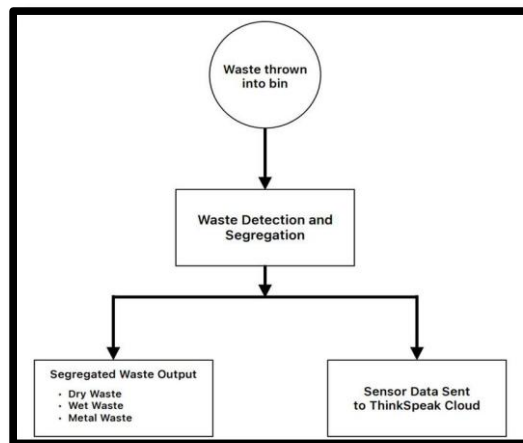


Fig 4.3.1 Data Flow Diagram- level 0

The above fig 4.3.1 shows the basic working of **Smart Waste Segregation System**. When the user throws the waste in the bin, the system is smart enough to identify the nature of the waste and then sends it through the segregation mechanism. The waste is then separated into the right category **dry, wet or metallic**. At the same time, the sensors detect the type and bin level of the waste and this information is transmitted to the ThingSpeak cloud via the microcontroller. This ensures proper segregation at the bin and also has real time sensor updates for monitoring via the dashboards.

4.3.2 Data Flow Diagram- LEVEL 1

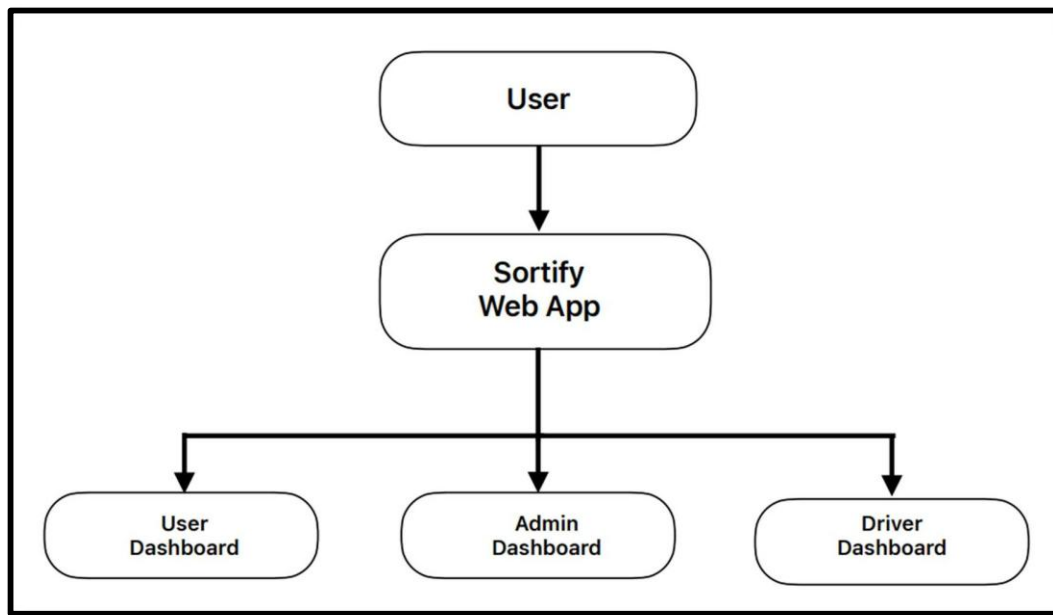


Fig 4.3.2 Data Flow Diagram- level 1

The above fig 4.3.2 is a representation of the interaction between the **Sortify Web App** and its three role based dashboards. The user enters the Sortify system via the main web application, which directs the user to the appropriate dashboard depending on his role. The **User Dashboard** can be used to allow residents to submit feedback, search segregation graphs, use the interactive map, and access the reward system and educational content. The **Admin Dashboard** allows administrators to access live bins level gauges via **ThingSpeak** and control driver accounts via **Supabase**. The **Driver Dashboard** gives the driver with bin level information from ThingSpeak and also provide SMS alerts received through the **GSM module**. Together, these dashboards try to ensure a smooth communication between the IoT system and all users of the platform.

4.3.3 Data Flow Diagram- LEVEL 2

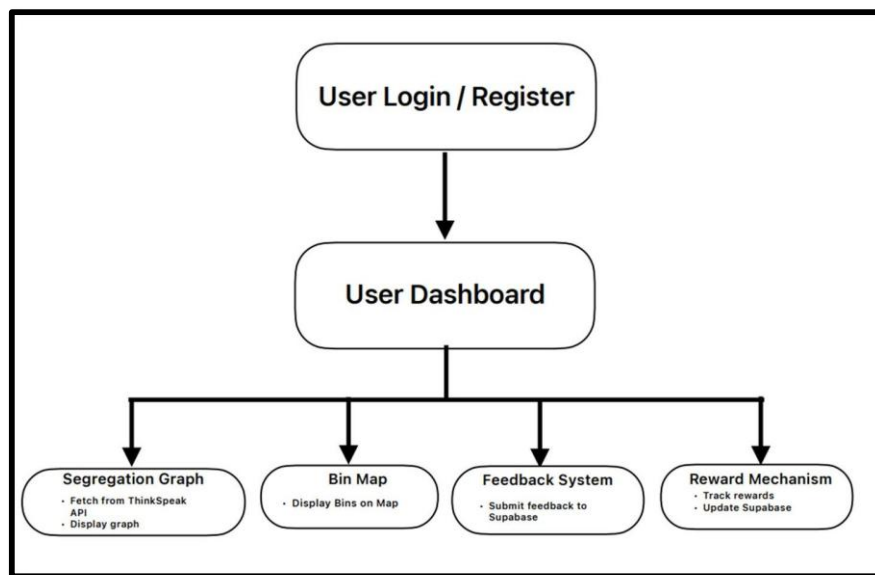


Fig 4.3.3 Data Flow Diagram- level 2

The above fig 4.3.3 represents the interaction between a user and the Sortify system through the User Dashboard. After logging in or registering, the user gets access to four main features. The Segregation Graph fetches in real time data of the type of waste generated from ThingSpeak and features it as an interactive graph. The Bin Map displays the available bins positions on an interactive map interface. Through the Feedback System, users can leave issues/comments which are stored in Supabase. The Reward Mechanism where users can track and update their reward progress which is also managed through Supabase. This flow demonstrates how the User Dashboard can give the consumers an easy interface to view the segregation insights, find the bins, share their feedback and interact with the reward system.

4.3.4 Data Flow Diagram- LEVEL 3

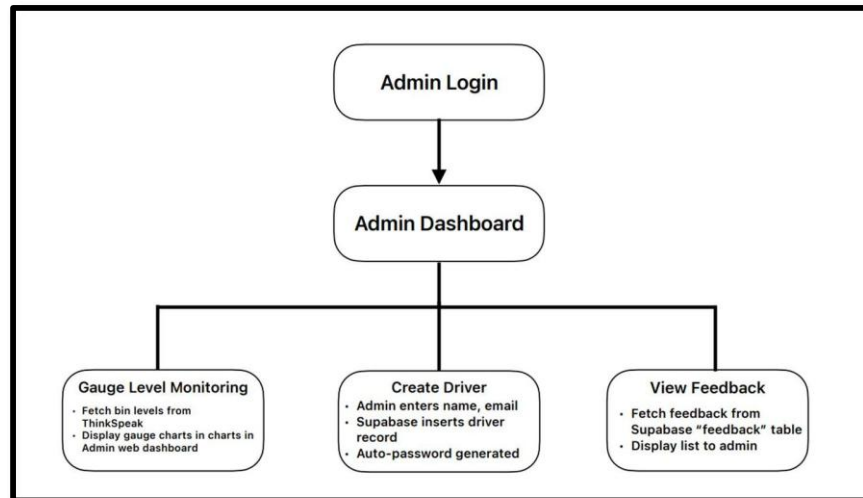


Fig 4.3.4 Data Flow Diagram- level 3

The above shown fig 4.3.4 represents Admin Dashboard. It demonstrates how the manage three major operations after logging in. First, the dashboard uses Thingspeak to get real time bin levels and use a gauge charts for easy monitoring. Second, the admin can now add new drivers by filling out simple information that is stored with Supabase including an auto-generated password. Finally the feedback from all users retrieved from the Supabase feedback table can be viewed by the admin. These processes make sure the bin status tracking, driver management, and feedback management is streamlined in one interface, making the admin dashboard efficient and easy to operate.

4.3.5 Data Flow Diagram- LEVEL 4

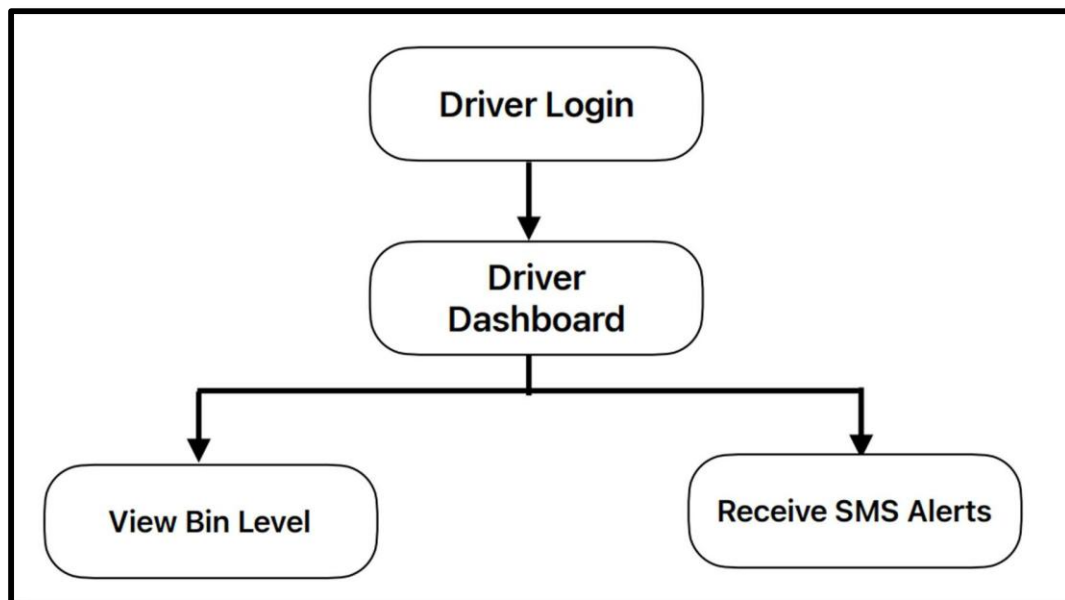


Fig 4.3.5 Data Flow Diagram- level 4

The above fig 4.3.5 shows process of the interaction of the driver after he/she has logged into the system. Once authenticated the driver can see the actual bin level real-time gauge, which is fetched from the ThingSpeak cloud via the web dashboard. This helps the driver keep track of the bins that are getting closer to being full. In addition, the driver gets SMS alerts which he receives straight from the GSM module whenever any bin reaches its threshold at its corresponding threshold, and hence, is immediately aware of it without strictly relaying on the dashboard. These two functions - bin level viewing and SMS alerts - assist the driver in quick response and coordination of bin collection.

4.4 Use Case Diagram

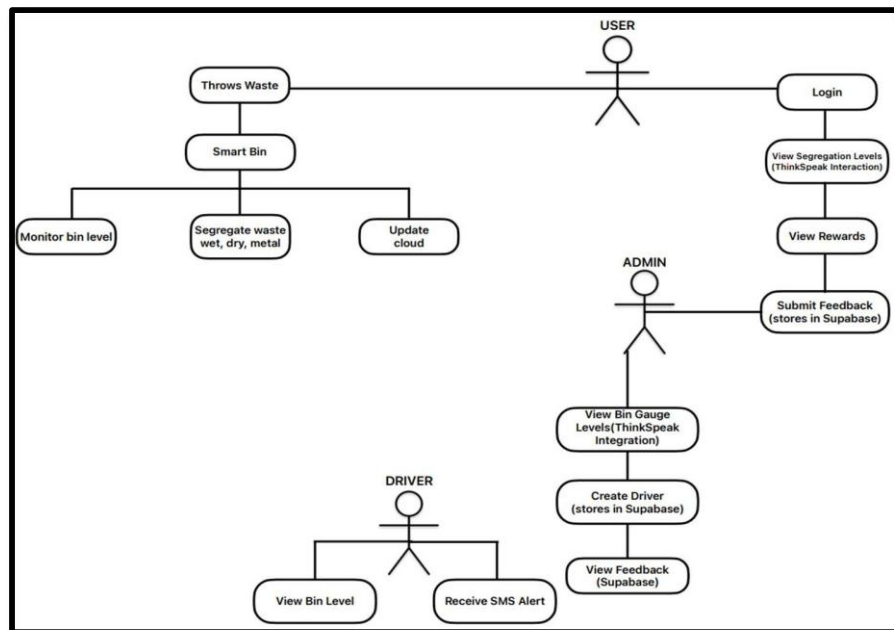


Fig 4.4 Use Case Diagram

The fig 4.4 for the Smart Bin Waste Management describes the interaction between the various actors - User, Admin, Driver and the Smart Bin in the Sortify smart waste segregation system.

The User triggers the process by disposing off waste inside the smart bin. The following actions are the major functionalities of the Smart Bin: Monitoring the level in the bin Taking action on waste, namely separating and identifying it as wet, dry or metallic by using sensors, which then will send the collected sensor data to the cloud via ThingSpeak.

On the web application side, the User is able to login and utilize a number of features under the User Dashboard. These involve viewing graphs for segregation based on data from ThingSpeak, viewing reward progress, and feedback as well as issues are stored in Supabase.

The Admin logs in through the Admin Dashboard and the three main things that are done is the checking of the levels of the bin gauge in real-time using ThingSpeak integration, creating the

driver accounts using Supabase and the feedback submitted by the users. This helps the administrators to check the condition of the bin and effectively deal with the system management

There are two ways the Driver communicates with the system. First the Driver Dashboard to see live bin levels that are obtained from ThingSpeak. Second, the driver receives SMS alerts through the GSM module whenever the bin is full and then action can be taken as quickly as possible without the need for complicated navigation systems.

Together, the aforementioned interactions demonstrate a complete process, from the Smart Bin handling the automatism of segregation and cloud updating, to the User and Admin dashboards that enable the participation of the community and the control of the administrator, to the Driver who makes sure that the collection of the waste is ensured in time through SMS and a live monitoring of the bins.

4.5 Class Diagram

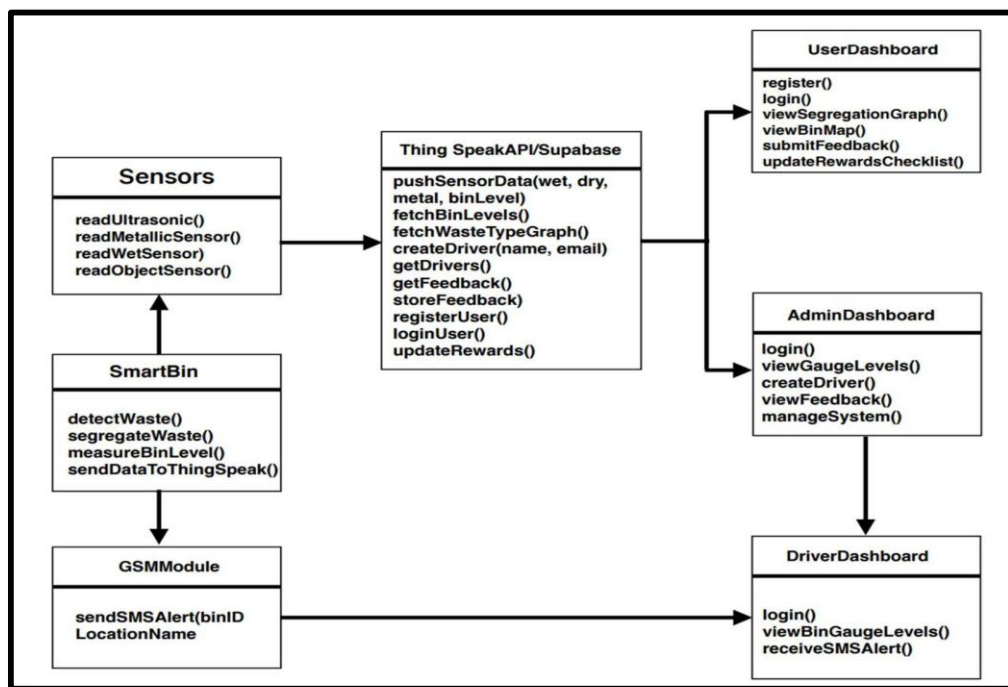


Fig 4.5 Class Diagram

This class diagram illustrates the organization and interactions of the Sortify smart waste segmentation system. The design begins from the Sensors class that encapsulates all the operations dealing with the sensors. It has the techniques of reading of ultrasonic, metallic, wetwaste and object detection sensors. These sensors are providing raw data for the SmartBin class.

The major functionality of the system is carried out in the class SmartBin. It detects the incoming waste, checks the type of waste with the help of the sensor inputs and switches on the segregation mechanism, calculates the bin fill level and sends the processed data to the cloud. When

the bin is full the SmartBin sends a message to the GSMModule class which provides for SMS alerts to the drivers by sending bin location and status messages.

The communication between the cloud is taken care of using ThingSpeakAPI/Supabase class. This class have methods to push data from the sensor to ThingSpeak, get bin levels and graphs of waste type ,create and manage function about the drivers,storing to get user feedback,registers and authenticates users, update reward mechanism This is the major connection between IoT module & web dashboards.

UserDashboard class provides functionality for a user to register, login, view segregation graphs, view the bin map, send feedback and update their reward checklist.

The DriverDashboard allows the drivers to login and have access to the bin level gages and also receive SMS alerts via the GSM module when the bins are full.

This architecture gives a smooth flow of data and operations between the sensors, cloud services, drivers, administrators, and the end users, resulting in a more efficient and responsive waste management solution.

4.6 Sequence Diagram

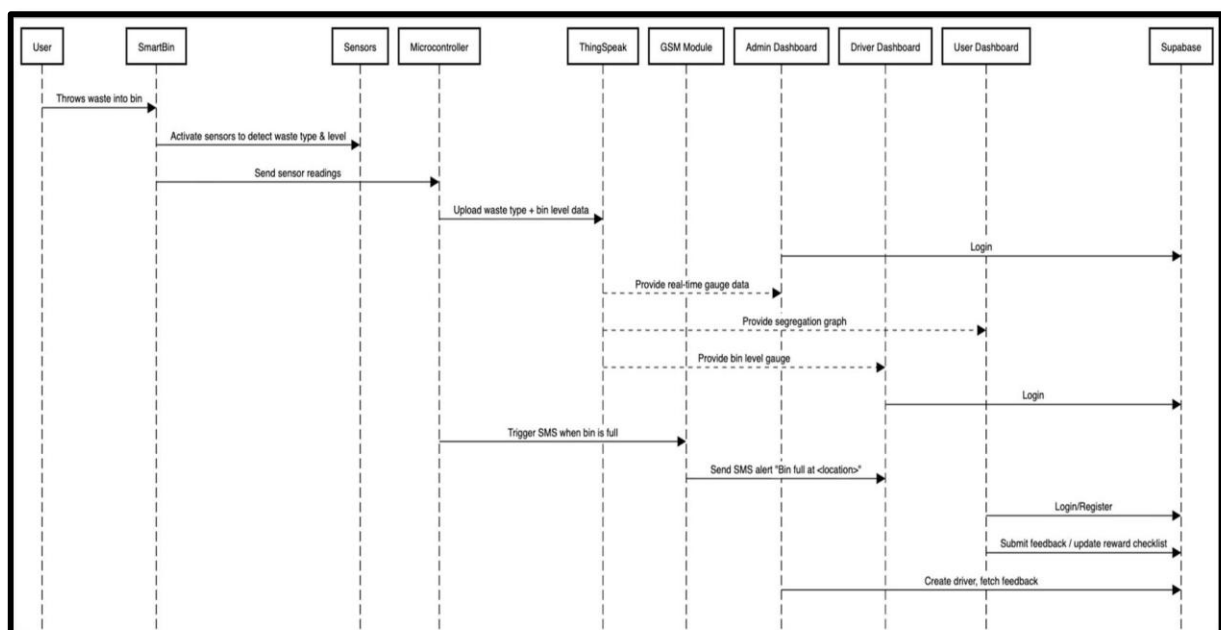


Fig 4.6 Sequence Diagram

The fig 4.6 of the Smart Waste Management System is the representation of the real-time interaction flow between the hardware components, cloud services, and the three dashboards of the Sortify system. The process starts when the User throws waste inside the Smart Bin. This is what activates the SmartBin, which in turn triggers its connected sensors that will act to identify the type of waste as well as the current bin fill level.

The Sensors take parameters such as metallic detection, level of wetness, presence of object and height of bin, and xcech this data to Microcontroller. The microcontroller uses these readings to upload the information about the type of waste and the level of waste in the bins to ThingSpeak, which provides the cloud visualization of real-time data.

Once the data has arrived at the ThingSpeak, it is available to the respective dashboards. The Admin Dashboard gets and shows the level of the gauge of the bin, and the User Dashboard fetches the segregation graphs according to ThingSpeak data. The Driver Dashboard also shows bin-level readings to monitor bin status for the drivers.

When the bin is full to a set level, the microcontroller controls the GSM Module that sends a SMS alert to the driver notifying him that the bin at a specific location is full. Meanwhile, Supabase uses all the web application interactions user login/registration, feedback submission, reward checklist update, and driver creation by the admin. All dashboards are interacting with Supabase to authenticate users and store application-related data.

4.7 Collaboration Diagram

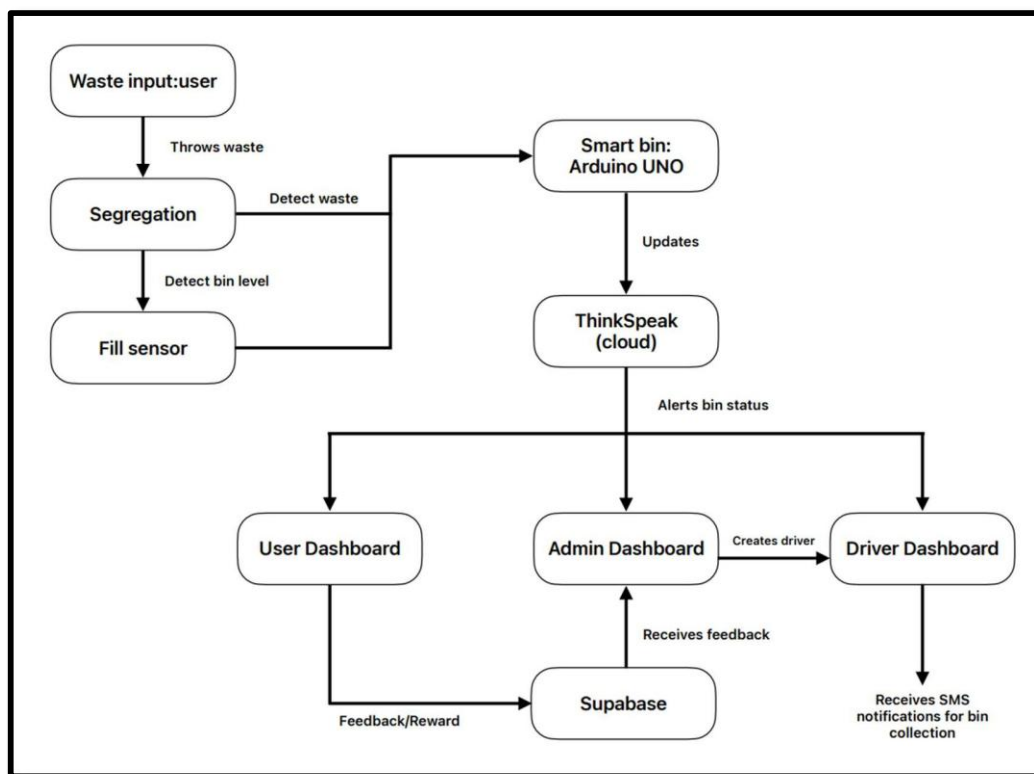


Fig 4.7 Collaboration Diagram

The fig 4.7 of the Smart Bin Waste Management System presents the hardware components, the cloud services and web dashboards interaction in the Sortify smart waste segregation system. The working process begins when the User puts waste in the bin. This action helps to activate the Segregation module - this determines what kind of garbage it is by applying a series of sensors and

sorts it out in the right category (wet copy, dry copy, metalware). Concurrently the Fill Sensor is measuring the amount of current in the bin.

Both the data types (waste and bin level) is passed to the Smart Bin (Arduino UNO) which process the gathered data and update the cloud. The Arduino sends this information to ThingSpeak, an IoT cloud platform for the storage and visualization of the real-time status of the bin. ThingSpeak then provides information about the bin status in real time to the Admin Dashboard, and to the Driver Dashboard so that the authorities and the drivers have immediate access to the bin status.

The Admin Dashboard is used to access the data that is stored on ThingSpeak at the gauge-level and uses Supabase to control system operations such as the creation of driver accounts and feedback submitted by users. The User Dashboard also provides a communication functionality with the Supabase for feedback submission and reward update, which can create an interesting interface for citizens to engage in responsible waste behaviour.

When the bin is full, Arduino triggers the GSM Module, which sends the direct SMS notification to the driver about pin point and status of bin. The Driver Dashboard is a complementary tool that displays the reading of selected gauges level in real time by ThingSpeak, allowing to check, in real time, the different conditions of the bins by the driver before collecting it.

Overall, the flow in the collaborative diagram results in a smooth and connected flow as follows: sensors(waste/ Bin Status) -> Arduino (ThingSpeak) -> Dashboards (Visualize data) -> Supabase (User/Admin Interaction)-> GSM Module (Alert Driver). This integrated structure provides for timely collection of waste, accurate monitoring and active involvement of user.

4.8 Activity Diagram

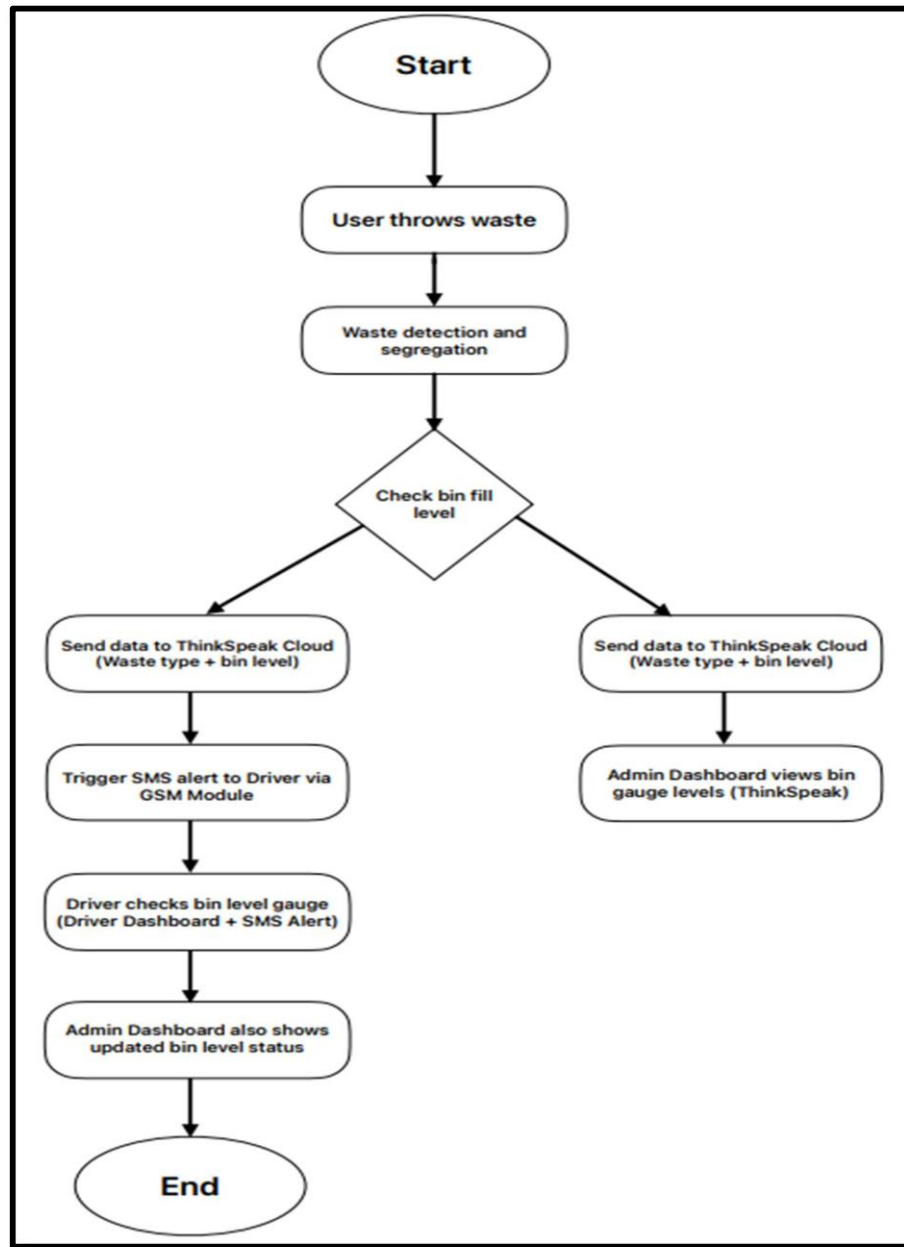


Fig 4.8 Activity Diagram

The fig 4.8 of Smart Waste Management System explains the complete work flow of Smart Waste Segregation System. The process starts when the user throws the waste into the smart bin. The system then conducts waste detection and segregation with help of its sensors. After segregation, the filling level of the bin is checked. No matter the level, the readings of the waste type and the bin level are sent on the ThingSpeak cloud. If the bin is full it sends an SMS alert to the driver via the GSM module so that he can see the bin level on his/her dashboard. At the same time, Admin Dashboard collects the updated bin level information from ThingSpeak making it a

continuous monitoring. This way enables the correct segregation, real-time data updates, and timely alerts for efficient waste management.

4.9 Modules

The Sortify system is structured in a modular system that collaborates to perform the essential work required in waste segregation, cloud communication, user interaction and the system's administrative monitoring. Each module is dedicated to a specific set of tasks which allows easier maintenance, scalability and clear separation of responsibilities.

4.9.1 Waste Segregation Module

This module handles the physical detection, classification, and segregation of waste inside the smart dustbin.

Functionality:

- Detects incoming waste using IR sensor
- Identifies waste type (wet/dry/metallic) using raindrop and metallic sensors
- Measures bin fill level using ultrasonic sensor
- Segregates waste using servo and stepper motors
- Sends real-time sensor data (waste type + bin level) to the cloud

Technology Used:

- Microcontroller (Arduino + sensor interfacing)
- Ultrasonic, IR, raindrop, and metallic proximity sensors
- Servo + stepper motors
- ThingSpeak API for IoT data upload
- GSM module for SMS alerts

Output:

- Properly segregated waste
- Real-time cloud updates
- SMS alert sent to driver when bin is full

4.9.2 Cloud Integration Module

This module manages all communication between the IoT bin and the cloud platforms.

Functionality:

- Pushes waste type and fill-level data to ThingSpeak
- Provides visual gauge charts and time-series graphs
- Stores user, driver, and feedback data in Supabase
- Ensures dashboards receive up-to-date data

Technology Used:

- ThingSpeak IoT Cloud
- Supabase (database + authentication)
- REST APIs

Output:

- Real-time bin readings available to dashboards
- Persistent storage for user/driver/feedback data

4.9.3 Admin Dashboard Module

This module provides the admin with tools to monitor bins and manage system operations.

Functionality:

- Admin login via Supabase
- View live bin-level gauges embedded from ThingSpeak
- Create drivers (auto-generate driver credentials through Supabase)
- View user feedback and system reports

Technology Used:

- Web application (HTML/CSS/JS/React)
- ThingSpeak visualizations
- Supabase authentication + database

Output:

- Updated bin monitoring interface
- Driver management
- Feedback management

4.9.4 User Dashboard Module

This module provides citizens with features to monitor segregation progress and interact with the system.

Functionality:

- User registration and login
- View segregation graphs (ThingSpeak data)
- View bins on an interactive map
- Submit issues and feedback to Supabase
- Interact with reward checklist system

Technology Used:

- Web application
- ThingSpeak graph embedding
- Supabase (auth + database)

Output:

- Graphical segregation insights
- Map-based bin location viewing
- Feedback stored for admin
- Updated reward progress

4.9.5 Driver Dashboard Module

This is a minimal module that supports driver notifications and status viewing.

Functionality:

- Driver login through Supabase
- View bin-level gauges from ThingSpeak
- Receive SMS notifications when bin is full

Technology Used:

- Web application
- ThingSpeak for bin data
- GSM module for SMS alerts

Output:

- Driver receives instant full-bin alert
- Driver can check live bin status

CHAPTER 5

IMPLEMENTATION

5.1 Objective 1: To design and deploy an IoT-based automated waste segregation system

The IoT segregation unit is implemented with the help of a microcontroller interfaced with the needed sensors i.e., ultrasonic sensor to detect the bin-level, the IR sensor is used to detect the object, raindrop sensor is used to detect the wet waste and metallic proximity is used to detect the metallic items. On input of waste, microcontroller categorizes the waste as wet, dry or metallic and actuates servo/stepper motors to rotate or open the appropriate waste compartment.

The microcontroller keeps sending the readings of the type of waste and the level in the bins to the ThingSpeak cloud by using the HTTP API request. When ultrasonic sensor found the bin is full, then it sends the SMS alert using SMS module (Apple and Android phone). This will ensure automated classification bins, updates to the cloud in real-time, as well as immediate notification when it needs to be collected.

5.2 Objective 2: To develop a user-friendly web application for reporting issues, viewing segregation insights, and promoting responsible waste behavior

The Sortify Web App has been designed using dedicated user, administrator, and driver dashboards. User Dashboard to register and log in using Supabase authentication. Users can view segregation graphs that are fetched from ThingSpeak, explore bin locations using an interactive map, submit issues or feedback to directly Supabase and interact with the reward progress checklist.

The Admin Dashboard meanwhile integrates ThingSpeak visualization so as to show realtime bin level gauge visualization charts. Would also be able to create driver accounts through Supabase & user submitted feedback. This design offers simplicity, efficiency and central management without unnecessary complexity.

The Driver Dashboard for drivers to login via Supabase and see the bin-level gauge from ThingSpeak and use the GSM module to notify them of the full bin via SMS.

5.3 Objective 3: To support environmental sustainability through data visualization, user engagement, and timely waste management alerts

Environmental sustainability is supported by enabling correct segregation at the source, providing residents with real-time awareness of waste patterns, and encouraging responsible habits through the reward mechanism. Users gain visibility into waste segregation trends through ThingSpeak graphs, while the smart bin automates sorting, reducing contamination and improving recycling potential.

The alert system is GSM in nature, which ensures that bins are cleared before overflowing, and surroundings are kept clean and unnecessary spillage of waste is avoided. The Admin Dashboard provides authorities with a means to be responsive to the reported feedback or issue from the community to enhance the quality of services and support cleaner environments.

5.4 Results and Discussions

The implementation indicates the working together of the IoT segregation unit and the cloud-connected web application. The smart dustbin is able to differentiate and identify the waste types (wet, dry, metals, etc.) and transmits data from the sensors to ThingSpeak in real-time, making all the dashboards thrive with the right information.

Users could register, report issues, view segregation graphs and use the reward system without difficulty. Admins monitored bin levels in ThingSpeak and managed drivers using Supabase and drivers got instant SMS alerts for full bin and also checked bin status on their dashboard.

Overall, the overall results confirm that integration of IoT automation with cloud visualization and easy web dashboards is an efficient way to improve the management of waste and community participation.

CHAPTER 6

TESTING

Testing was conducted for proper functionality of IoT segregation unit, cloud communication, and the three dashboards of the Sortify system. All the components were tested under real conditions with a real smart dustbin prototype with the help of sensors, a microcontroller hardware, wireless modules and live cloud integration.

6.1 Test Environment

The Smart Bin prototype was constructed with the help of an Arduino UNO connected to ultrasonic sensors, metallic- waste, (proximity switch) wet waste (rain drop), dry waste (IR object detection) detecting sensors. GSM module was attached for SMS alert. The entire system was powered using a steady 5V supply.

Waste Segregation & Sensor Monitoring

Each sensor was tested individually and then collectively:

- **Wet waste detection** was tested using the **raindrop sensor**, which successfully identified moisture when damp organic waste was placed near the sensing surface.
- **Metallic waste detection** was validated using the **proximity switch sensor**, which correctly triggered when metallic objects were introduced.
- **Dry waste detection** was validated using the **IR object detection sensor**. When the IR sensor turned TRUE while the raindrop sensor and proximity switch remained FALSE, the system correctly classified the waste as dry.
- **Ultrasonic sensor** was tested by placing waste at different heights inside the bin to verify accurate **fill-level** measurement.

Cloud Connectivity (ThingSpeak)

The microcontroller was connected to Wi-Fi to upload:

- Waste type (wet/dry/metal)
- Bin fill level values.

ThingSpeak channels were monitored to ensure real-time updates and accurate data plotting.

GSM Alert System

The GSM module was tested by filling the bin to the predefined threshold. The module successfully triggered:

- SMS message
- Delivery to driver's registered mobile number

Dashboard Testing (Web App)

- Admin Dashboard pulled gauge readings directly from ThingSpeak.
- User Dashboard displayed segregation graphs via ThingSpeak API.
- Driver Dashboard fetched bin-level updates and received SMS alerts.
- Supabase was used for login, feedback storage, reward updates, and driver creation.

6.2 Unit Testing

Unit testing was performed on each hardware and software component individually to ensure that every module of the system operates correctly before integration.

Table 6.2 Unit test cases

Step	Test Data / Component	Expected Result	Observed Result	Remarks
Step 1	Ultrasonic Sensor	Correctly measure bin height and produce accurate fill-level values	Stable and precise readings at different heights	Pass
Step 2	Raindrop Sensor (Wet Waste)	Detect moisture and classify waste as wet	Correctly detected damp/wet waste samples	Pass
Step 3	Proximity Switch (Metal Waste)	Detect metallic objects reliably	Triggered correctly for all metal samples	Pass
Step 4	IR Object Sensor (Dry Waste)	Detect object presence; classify dry waste when other sensors = false	Detected dry waste accurately with correct Boolean logic	Pass
Step 5	Segregation Mechanism (Servo + Stepper)	Move flap/rotate bin to correct compartment based on waste type	Motors activated correctly and routed waste accurately	Pass
Step 6	GSM module SMS Alerts	Send SMS when bin reaches full capacity	SMS delivered instantly to driver	Pass
Step 7	GSM module HTTP Upload → ThingSpeak	Send waste type + fill-level data to ThingSpeak in real time	ThingSpeak channel updated immediately	Pass
Step 8	Supabase Auth	Validate login/registration for User/Admin/Driver	All roles authenticated successfully	Pass
Step 9	Supabase Database	Store feedback, reward status, driver data	All entries created/retrieved without errors	Pass

6.3 Integration Testing

Integration testing was performed in order to ensure smooth functioning between all interconnected components of the smart waste segregation system. In continuous working flows, the sensors, microcontroller, GSM module, Thing Speak cloud, and the three dashboards were tested in a combination manner. Waste was inserted into the bin and the integrated functioning of the raindrop sensor, proximity switch, IR sensor, and ultrasonic sensor was checked as data got sent to the microcontroller. The GSM module was tested for uploading real-time values to ThingSpeak using mobile data as well as for triggering an SMS alert when the bin is full. Dashboards were validated to ensure that the Admin Dashboard in Segregation Graphs from the measurements which were retrieved from ThingSpeak, the User Dashboard displayed the correct segregation values, and the Driver Dashboard displayed updated bin status. Supabase integration was done for authentication, feedback storage, reward updates and driver creation for all data exchanges to take place properly between all modules. Overall, the integrated system proved to be reliable in hardware, cloud, and web interface coordination.

6.4 System Testing

System testing was conducted to test the full end-to-end functionality of the smart waste management solution in real working conditions. The complete workflow from waste insertion to segregation to updating of cloud to Sms and dashboard visualization was tested as a unified one. The smart bin was able to correctly classify the waste into categories of wet, dry, and metallic and update the fill levels in real-time. The GSM module was able to send the sensor readings to ThingSpeak and send full bin alerts to the driver. Dashboards were accurate and real-time and Supabase operations such as login, feedback submission, rewards tracking, and driver management went smoothly. The system reacted appropriately to all of the user touch, hardware events and triggers from the cloud, and confirmed that the full architecture worked seamlessly as one integrated application.

6.5 Functional Testing

Functional testing confirmed that all system features met the required specifications. The automatic waste detection and segregation operated correctly based on sensor logic, and bin-level monitoring stayed in sync with real-time readings. The GSM module reliably uploaded data to ThingSpeak and sent SMS alerts for full-bin conditions. The Admin Dashboard displayed live levels and supported driver management, while the User Dashboard enabled login, segregation graph viewing, reward updates, and feedback submission. The Driver Dashboard accurately showed bin levels and aligned with SMS notifications. Overall, all functions performed as designed, validating the system's goals of automation, cloud visualization, and role-based accessibility.

CHAPTER 7

RESULTS



Fig 7.1 Waste Segregation System – Hardware Prototype

The above Fig. 7.1 illustrates the complete hardware prototype of the IoT-based Waste Segregation System. The setup showcases the automated sorting mechanism that segregates incoming waste into three categories: dry, wet, and metal. The prototype integrates an ultrasonic sensor module and a servo-motor-based rotating mechanism to detect and direct waste into the respective bins. Each bin is clearly labelled to indicate its designated waste type. Additionally, the system incorporates a GSM module mounted on the board, which continuously monitors bin levels and automatically sends SMS alerts to the concerned authority whenever any bin reaches its full capacity.



Fig 7.2 Metal Waste Detection Using Proximity Sensor

Fig. 7.2 shows the system detecting metal waste using a proximity switch. When the IR sensor senses an incoming object, the proximity sensor checks if it is metallic. In this test scenario, the metal object is correctly identified, and the system prepares to rotate the mechanism to direct it into the metal bin. The red illumination indicates that the sensing and sorting process is active.



Fig 7.3 Wet Waste Detection Using Rain Drop Sensor

Fig. 7.3 illustrates the detection of wet waste using the rain drop sensor. After the IR sensor detects an object, the rain sensor identifies moisture content and classifies it as wet waste. The system then rotates the mechanism to drop the waste into the wet bin.



Fig 7.4 Dry Waste Detection When Metal & Wet Sensors Are False

Fig. 7.4 shows the sorting process for dry waste. When the IR sensor detects an object but both the proximity sensor (metal detection) and rain sensor (wet detection) give false outputs, the system classifies the object as dry waste and directs it to the dry bin.



Fig 7.5 Final Output – Segregated Waste in Individual Bins

Fig. 7.5 shows the final output of the waste segregation system, where metal, wet, and dry waste have been successfully sorted into their respective bins. Each sensing module-IR, proximity switch,

and rain drop sensor—worked together to ensure accurate classification, demonstrating the complete functionality of the automated waste segregation model.

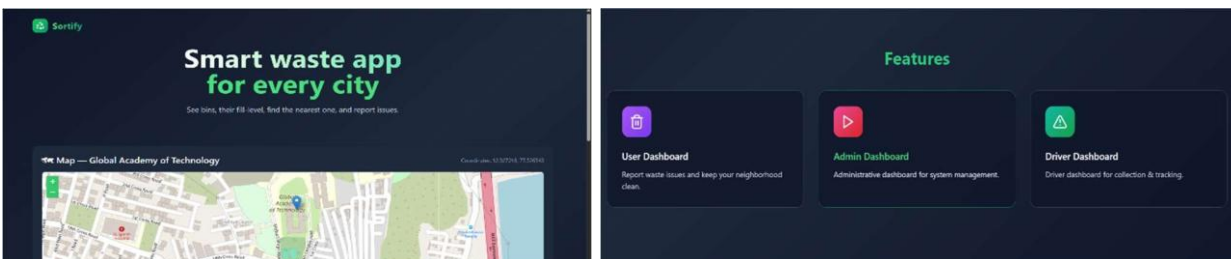


Fig 7.6 Sortify Web Application – Landing Page and Key Features

Fig. 7.6 shows the landing page of the Sortify smart waste management web application. The interface includes a real-time interactive map that displays bin locations around the Global Academy of Technology, allowing users to quickly view bin positions and their fill levels. The page also highlights the main features of the platform, including access to the User, Admin, and Driver dashboards, enabling efficient monitoring, management, and collection planning within the system.

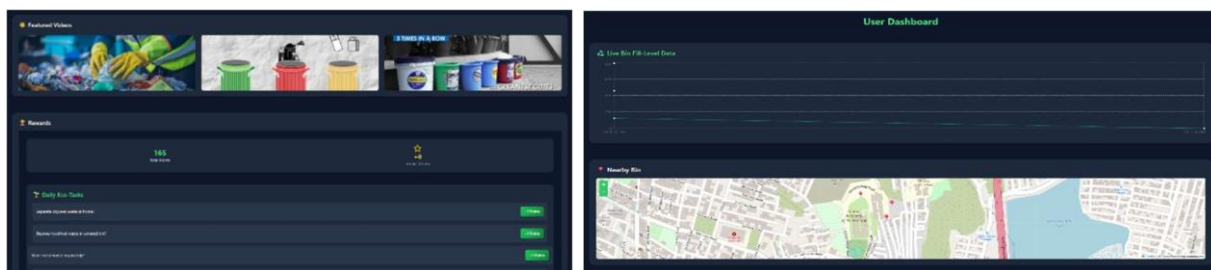


Fig 7.7 User Dashboard of the Sortify Web Application

Fig. 7.7 represents the User Dashboard of the Sortify web application. It displays a live bin-level graph that updates in real time using data fetched from the ThingSpeak IoT cloud. The dashboard also includes an interactive map showing the locations of the three smart bins deployed in the Global Academy of Technology. Additionally, it provides educational YouTube-linked featured videos, a reward and streak system that motivates users to complete daily eco-friendly tasks, and a dedicated feedback and report section that forwards user submissions directly to the admin through the Supabase cloud backend. This dashboard is designed to promote user engagement while ensuring smooth communication within the waste management system.

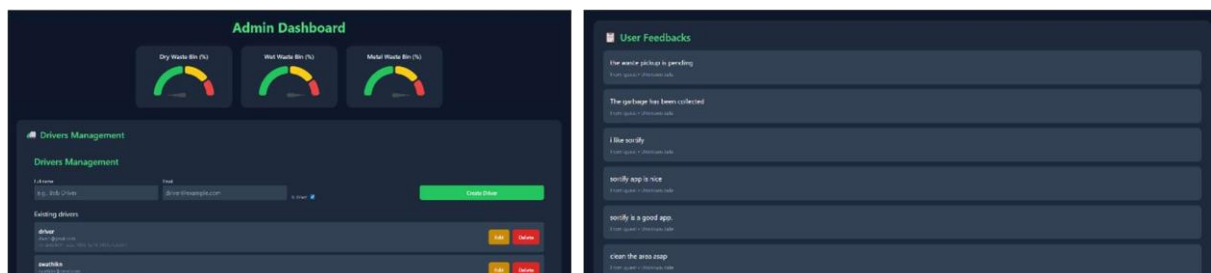


Fig 7.8 Admin Dashboard of the Sortify Web Application

Fig. 7.8 shows the Admin Dashboard, which provides real-time monitoring of the dry, wet, and metal waste bins using live fill-level gauges connected to the ThingSpeak IoT cloud. The dashboard also includes a driver management section, allowing the admin to create new driver accounts with automatically generated temporary login keys. Additionally, all user-submitted feedback and reports are displayed in the feedback panel, stored and managed through the Supabase cloud backend. This dashboard enables efficient supervision, driver coordination, and system management.

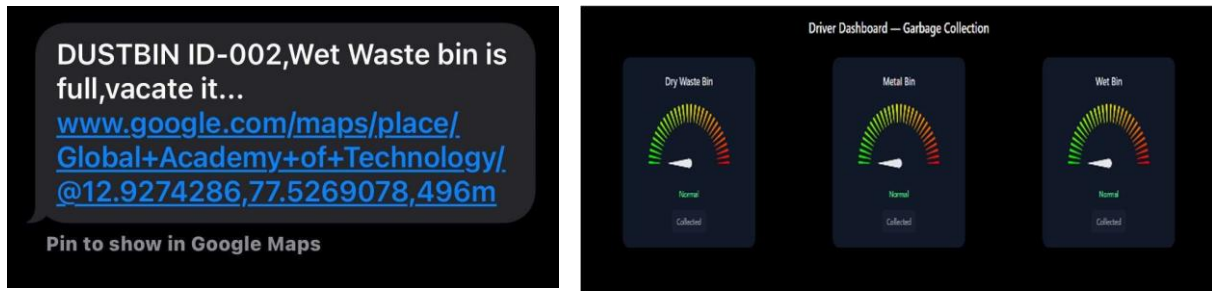


Fig 7.9 Driver Dashboard With GSM Alert Integration

Fig. 7.9 illustrates the Driver Dashboard of the Sortify system, which displays the live fill levels of the dry, metal, and wet waste bins through gauge indicators. These values are continuously updated using the ultrasonic sensor readings sent to the ThingSpeak IoT cloud. When any bin becomes full, the GSM module automatically sends an SMS alert to the driver, containing the Dustbin ID, the type of waste that is full, and a Google Maps location link for quick access. After collecting the waste, the driver can log in to the web application and confirm collection by clicking the “Collected” button. This seamless integration ensures efficient waste pickup and real-time tracking of bin status.

CONCLUSION

The Sortify system is a successful solution that combines an IoT-based waste segregation unit with a web application connected to a cloud for better efficiency and engagement of waste management. The smart dustbin precisely sorting waste items into wet, dry and metallic categories through the use of sensors and mechanical actuation, and continuously updating the data of bins and types of waste to the ThingSpeak cloud. This allows for the bin status to be seen in real-time by Admin, User and Driver dashboards.

Through the web application, users will be able to register and view segregation trends, access an interactive map of the bins, provide feedback, and engage in a reward-based system. The Admin Dashboard allows the authorities to control live bin levels and driver accounts via Supabase. Drivers receive an instant SMS alert on the GSM module in a bin is full and take timely action to prohibit the bin from overflowing.

Overall, the project demonstrates how IoT automation, cloud visualization, and lightweight dashboards can work together to improve urban cleanliness, encourage responsible waste practices, and provide a scalable found

BIBLIOGRAPHY

[1] S. Mehta and A. Rathour, "Innovative Waste Management: The Efficiency of IoT-Enabled Smart Bins," 2024 7th International Conference on Circuit Power and Computing Technologies (ICCPCT), Kollam, India, 2024, pp. 467-471, doi: 10.1109/ICCPCT61902.2024.10672653.

keywords: {Waste management;Costs;Urban areas;Systems architecture;Carbon dioxide;Real-time systems;Software;Smart Cities;Waste Management;Internet of Things (IoT);Ultrasonic Sensors;Embedded Devices;Urban Sanitation},

[2] M. P. Varghese, V. S. Anooja, R. Akhila, M. Krishnakumar and A. Xavier, "IoT-Based Smart Waste Management System with Level Indicators for Effective Garbage Waste Segregation," 2024 Third International Conference on Distributed Computing and Electrical Circuits and Electronics (ICDCECE), Ballari, India, 2024, pp. 1-5, doi: 10.1109/ICDCECE60827.2024.10548589.

keywords: {Waste management;Visualization;Urban areas;Recycling;Internet of Things;Sustainable development;Waste disposal;Garbage Segregation;Internet of Things;Level Indicators;Smart Waste Management;Urban areas},

[3] M. P. J. E. G S and L. R. B. Raj S, "Garbage monitoring system using IoT," 2024 2nd International Conference on Networking and Communications (ICNWC), Chennai, India, 2024, pp. 1-6, doi: 10.1109/ICNWC60771.2024.10537580. keywords: {Waste management;Wireless sensor networks;Waste materials;Data analysis;Wireless networks;Real-time systems;Servers;Industrial IoT device;actual-time systems;systems of sensors;distant observation;solid refuse management;dust bin;solar-powered},

[4] M. V. Rajesh, A. Lakshmanarao, J. S. A. Rongala and C. D. Priyanka, "IOT based smart system for garbage detection and segregation," 2023 IEEE Silchar Subsection Conference (SILCON), Silchar, India, 2023, pp. 1-4, doi: 10.1109/SILCON59133.2023.10405323. keywords: {Economics;Waste management;Urban areas;Robot sensing systems; Acoustics; Recycling; Monitoring;Garbage;IOT;Smart Dustbin;Ultrasonic Sensors},

[5] R. Tandon and A. Bansal, "IoT-enabled Smart Waste Management: Leveraging the Power of IoT for Sustainable Solutions," 2023 3rd International Conference on Innovative Sustainable Computational Technologies (CISCT), Dehradun, India, 2023, pp. 1-6, doi: 10.1109/CISCT57197.2023.10351210. keywords: {Waste management;Wireless sensor networks;Costs;Sociology;Transportation;Electronic waste;Internet of Things;Internet of things (IoT);Waste Management;Smart Cities;Computer vision;Sustainable development}

APPENDIX

SDG 11: Sustainable Cities and Communities



Our project aligns with **SDG 11: Sustainable Cities and Communities** by improving cleanliness, hygiene, and waste handling in public spaces through smart automation. Traditional waste management systems rely heavily on manual monitoring, which often leads to overflowing bins, unhygienic surroundings, and inefficient collection. By automating waste segregation and continuously monitoring bin levels, Sortify reduces visual pollution and helps maintain cleaner, more liveable cities. Real-time sensor data is sent to the cloud, and drivers are notified immediately when bins are full, ensuring timely collection, preventing garbage pile-up, reducing health hazards, and avoiding unnecessary collection trips. In addition, smart dashboards provide authorities with clear insights into waste conditions, enabling informed decision-making and efficient resource planning. Sortify also promotes citizen awareness and participation by allowing users to track segregation trends, locate bins, report issues, and engage in eco-friendly reward-based activities, fostering community responsibility and contributing to smarter, safer, and more sustainable urban environments.

Swathi K N 1GA22CS174

review paper - Copy.doc

 Global Academy of Technology, Bengaluru

Document Details

Submission ID**trn:oid::3618:120651353****Submission Date****Nov 10, 2025, 10:57 AM GMT+5:30****Download Date****Nov 10, 2025, 10:59 AM GMT+5:30****File Name****review paper - Copy.doc****File Size****98.0 KB****7 Pages****3,627 Words****23,872 Characters**

4% Overall Similarity

The combined total of all matches, including overlapping sources, for each database.

Filtered from the Report

- Bibliography
- Quoted Text
- Cited Text
- Small Matches (less than 8 words)

Match Groups

-  **13 Not Cited or Quoted 4%**
Matches with neither in-text citation nor quotation marks
-  **0 Missing Quotations 0%**
Matches that are still very similar to source material
-  **0 Missing Citation 0%**
Matches that have quotation marks, but no in-text citation
-  **0 Cited and Quoted 0%**
Matches with in-text citation present, but no quotation marks

Top Sources

- 3%  Internet sources
- 4%  Publications
- 0%  Submitted works (Student Papers)

Integrity Flags

0 Integrity Flags for Review

No suspicious text manipulations found.

Our system's algorithms look deeply at a document for any inconsistencies that would set it apart from a normal submission. If we notice something strange, we flag it for you to review.

A Flag is not necessarily an indicator of a problem. However, we'd recommend you focus your attention there for further review.

Match Groups

-  **13 Not Cited or Quoted** 4%
Matches with neither in-text citation nor quotation marks
-  **0 Missing Quotations** 0%
Matches that are still very similar to source material
-  **0 Missing Citation** 0%
Matches that have quotation marks, but no in-text citation
-  **0 Cited and Quoted** 0%
Matches with in-text citation present, but no quotation marks

Top Sources

- 3%  Internet sources
- 4%  Publications
- 0%  Submitted works (Student Papers)

Top Sources

The sources with the highest number of matches within the submission. Overlapping sources will not be displayed.

- 1** Internet
turcomat.org 1%
- 2** Publication
"Advances in Artificial Intelligence and Machine Learning", Springer Science and ... <1%
- 3** Internet
journals.tultech.eu <1%
- 4** Publication
A R Vaishnavi, Sahana G, N Guruprasad. "Wearable Devices in the IoT: A Security ... <1%
- 5** Publication
V. Sharmila, S. Kannadhasan, A. Rajiv Kannan, P. Sivakumar, V. Vennila. "Challeng... <1%
- 6** Internet
www.isca.co.in <1%
- 7** Publication
Abdelaziz Daas, Bilal Sari, Fouzi Semchedine, Mourad Amad. "Optimizing waste m... <1%
- 8** Publication
Inam Ullah, Inam Ullah Khan, Mariya Ouaisa, Mariyam Ouaisa, Salma El Hajjami... <1%
- 9** Internet
bpasjournals.com <1%
- 10** Internet
www.ieindia.org <1%

Swathi K N 1GA22CS174

review paper - Copy.doc

 Global Academy of Technology, Bengaluru

Document Details

Submission ID

trn:oid:::3618:120651353

Submission Date

Nov 10, 2025, 10:57 AM GMT+5:30

Download Date

Nov 10, 2025, 10:59 AM GMT+5:30

File Name

review paper - Copy.doc

File Size

98.0 KB

7 Pages

3,627 Words

23,872 Characters

*% detected as AI

AI detection includes the possibility of false positives. Although some text in this submission is likely AI generated, scores below the 20% threshold are not surfaced because they have a higher likelihood of false positives.

Caution: Review required.

It is essential to understand the limitations of AI detection before making decisions about a student's work. We encourage you to learn more about Turnitin's AI detection capabilities before using the tool.

Disclaimer

Our AI writing assessment is designed to help educators identify text that might be prepared by a generative AI tool. Our AI writing assessment may not always be accurate (i.e., our AI models may produce either false positive results or false negative results), so it should not be used as the sole basis for adverse actions against a student. It takes further scrutiny and human judgment in conjunction with an organization's application of its specific academic policies to determine whether any academic misconduct has occurred.

Frequently Asked Questions

How should I interpret Turnitin's AI writing percentage and false positives?

The percentage shown in the AI writing report is the amount of qualifying text within the submission that Turnitin's AI writing detection model determines was either likely AI-generated text from a large-language model or likely AI-generated text that was likely revised using an AI paraphrase tool or word spinner.

False positives (incorrectly flagging human-written text as AI-generated) are a possibility in AI models.

AI detection scores under 20%, which we do not surface in new reports, have a higher likelihood of false positives. To reduce the likelihood of misinterpretation, no score or highlights are attributed and are indicated with an asterisk in the report (*%).

The AI writing percentage should not be the sole basis to determine whether misconduct has occurred. The reviewer/instructor should use the percentage as a means to start a formative conversation with their student and/or use it to examine the submitted assignment in accordance with their school's policies.

What does 'qualifying text' mean?

Our model only processes qualifying text in the form of long-form writing. Long-form writing means individual sentences contained in paragraphs that make up a longer piece of written work, such as an essay, a dissertation, or an article, etc. Qualifying text that has been determined to be likely AI-generated will be highlighted in cyan in the submission, and likely AI-generated and then likely AI-paraphrased will be highlighted purple.

Non-qualifying text, such as bullet points, annotated bibliographies, etc., will not be processed and can create disparity between the submission highlights and the percentage shown.



A Survey on Smart Waste Management Systems: Trends, Challenges, and Future Directions

Prof. Shwetha N
Dept. of Computer Science and
Engineering
Global Academy of Technology
Bengaluru, India
shwetha.n@gat.ac.in

Swara Shetty M
Dept. of Computer Science and
Engineering
Global Academy of Technology
Bengaluru, India
swarshtt@gmail.com

Harshitha N K
Dept. of Computer Science and
Engineering
Global Academy of Technology
Bengaluru, India
nkharshitha19@gmail.com

Swathi K N
Dept. of Computer Science and
Engineering
Global Academy of Technology
Bengaluru, India
swathikn6142@gmail.com

Shreya N
Dept. of Computer Science and
Engineering
Global Academy of Technology
Bengaluru, India
shreyasana3@gmail.com

Abstract—The increase in urban population has multiplied exponentially the problems in solid waste management including improper collection, poor segregation and environmental hazards. The established practices lack continuous monitoring and upgrade. Now, waste fill-level sensing, automatic triggered alerts and analytics-based scheduling can be provided through IoT enabled smart dustbins features [1]. Comprehensive incentive based arguments like Credibin [2] and fairness-driven modes [4] have been proposed. Other papers suggest a cost-effective IoT monitoring [5], level indicators waste separation [6], and routing [7]. This field is further extended by the sensor-based sorting [8], the IoT-enabled sustainable sorting systems [9], and the integrated smart dustbin [10]. This contribution is a compilation of these contributions and exposes gaps, including the lack of a common framework that unifies all these contributions, including those focused on segregation, real-time monitoring, routing optimisation and citizen engagement. To overcome these negative aspects, we propose a solution by implementing a next generation of Sortify, which is defined by the integration of sensor-based smart bins, cloud connectivity, GPS-enabled collection, and reward mechanisms based on the communities. Assessing the impact that waste management practices have in such settings, we talk about a structural framework that envisions a holistic and scalable waste management model for smart cities that is environmentally sustainable and promotes citizen engagement.

Keywords:(Smart Waste Management, IoT-Enabled Bins, Waste Segregation, Sensor Networks, Route Optimization, Cloud Computing, Incentive Mechanisms, Community Participation, Urban Sustainability)

I. INTRODUCTION

The explosion in urban population growth, the changing lifestyles and the accelerating rate of industrialization have exerted a high degree of pressure on municipal solid waste (MSW) management systems worldwide. Conventional methods, which rely largely on manual monitoring and fixed waste collection times, often cause problems in the process (such as overflowing the bins, inefficient routing, low waste segregation, risking the environment). These limitations point to the important need of sustainable frameworks that support data-driven and citizen-inclusive waste management that aligns with the goals of smart cities of the future.

Recent technologies in the Internet of Things (IoT) have paved the way for the development of intelligent waste management solutions, involving the use of sensors, cloud computing, and mobile technologies. Smart bins with ultrasonic sensors, infrared sensors, or weight sensors can help in monitoring the fill level in real-time and optimize the collection schedule and save costs [1]. To increase engagement of users, some innovative systems such as Credibin have included reward-based disposal systems to incentivize sustainable disposal practices in campus communities [2]. Additional developments consist of SmartBin+, which uses digital incentives to encourage responsible waste disposal [11], an implementation of radio frequency identification (RFID) enabled smart bins which automate the tracking of litigation [15], and gamification strategies aimed at boosting user participation in sustainably based activities [12].

Concurrently, scheduling and optimization methodologies have become critical in order to increase the efficiency of waste collection. Decision-support systems help municipalities to plan for optimized collection routes [3], or the fairness-oriented design helps to ensure equal distribution of waste collection in different regions [4]. Other studies use the optimized routing techniques to optimize fuel consumption and emissions [7]. Furthermore, low-cost prototypes connected to the Internet (IoT) have been constructed for easily deployable pilot projects on a large scale in urban and rural areas [5], [14], while bin-level segregation and fill-indicator systems are intended to enable improved waste segregation at the waste origin [6]. Emerging technologies like machine learning are further adding to these capabilities through intelligent classifications of waste and predictive waste monitoring such as in the SentiTrash framework [13]. There are also comprehensive reviews on the transformative potential of IoT for the establishment of sustainable waste management ecosystems [9], [10].

Although these contributions are promising, most of the research efforts tend to focus on the waste management challenges in a fragmented manner, focusing individually on aspects such as monitoring [5], route optimization [7] or incentive mechanisms [2], [11], [12], [15]. Only a few studies attempt to combine segregation, optimization, incentives and the involvement of citizens in a forward-

looking, integrated effort. This fragmentation shows us that there is a major gap in research to develop solutions that are not only technologically effective, but also inclusive in terms of the social power dynamics, as well as sustainable for the environment.

This review paper discusses the latest development in the IoT-enabled Waste management systems, which emphasized particularly on some themes such as monitoring, incentive mechanisms, optimization, segregation, and integration frameworks. Anchored by the fundamental study "Innovative Waste Management: The Efficiency of IoT-Enabled Smart Bins" [1], the paper evaluates some significant progresses and existing insufficiencies within the literature. Furthermore, it introduces Sortify as a representative framework which embodies some of the characteristics of the next generation discussed. The general goal is to highlight the development towards holistic, sensor-based and societal-centered waste management solutions.

II. LITERATURE REVIEW

Research on smart waste management systems and IoT-enabled solutions is being used in various domains through analysis of real-time monitoring systems and incentive-based, optimized routing methods, waste segregation as well as people participation studies. The research paper reviews fifteen exemplary studies, which have provided the general principles for the establishment of modern waste management systems.

Mehta et al. [1] developed a research work based on Innovative Waste Management: The Efficiency of IoT-Enabled Smart Bins, where IoT-enabled smart bins were utilized to monitor the bin fill level and also to give real-time alerts. The findings demonstrated improvement of the collection performance and a reduction in human involvement for the tasks. Most research works have been conducted on the monitoring system instead of developing the reward system or the sophisticated routing schemes.

Abishek et. [2] came up with an IoT-based smart waste disposal system named Credibin which works in a campus environment. The system combined smart bins and a reward system to get people to manage their waste appropriately. While the approach was effective at modifying behaviour, its use was limited to the institutional environments in which control was retained.

Batoon et. An optimization framework and decision support tools have been developed by [3] to schedule waste collection operations. The proposed system was able to improve the performance of waste vehicles in terms of decision-making support algorithms, and reduce collection costs. The research activities were focused on development of algorithms, but did not include real-time monitoring systems or incentive elements in the system.

Rahman et al. [4] developed an Internet of Things (IoT) based smart system for the management of garbage management system using a fair resource allocation system. The research was instrumental in setting up an even distribution of collection resources within urban zones resulting in a reduction of existing gaps in relation to quality of service. The system lacked routing optimisation and models to support citizen participation.

Mithun et al. [5] proposed a low-cost IoT-based tracking system for monitoring the garbage levels that detects the basic level of garbage and can be extended to urban areas. The system has simple waste management functions and is relatively cheap, but there are no waste separation functions and operation control systems.

Varghese et al. [6] proposed an IoT-based smart waste management system in which level indicators are used for proper segregation of waste at source. This originated sustainable disposal measures; however it was limited by the lack of routing optimization and incentive driven methods.

Dharan et al. [7] presented a system of smart waste management with optimized routing algorithm. By using data from IoT sensors for forming collection routes, their solution resulted in savings in fuel and enhanced the temporal efficiency. However, the study was based mainly on routing optimization, with no consideration for implementation with incentive programs or waste segregation.

Rajesh et al. [8] introduced an IoT-based waste detection model for waste segregation using automated waste classification with the help of sensor technology for better source level segregation. However, their research did not go as far as municipal system integration or encouraging community participation.

In a substantial work presented by Tandon et al. [9], IoT enabled waste management solutions for sustainability have been reviewed in detail. Their work highlighted the transformational possibilities of IoT in smart city environments emphasizing mostly conceptual models. On the other hand, it had no practical applications or system design.

Rupesh et al. [10] presented IoT enabled smart dustbin system for urban waste management allowing real-time monitoring the fill levels. Although this function was implemented, the system was limited to basic functioning and did not include optimization strategies and incentive schemes.

Chavan et al. [11] proposed IoT-based waste segregation using inductive proximity, moisture and ultrasonic sensor that could sort waste into wet, dry and metal waste. Although the system has been proven to be effective in terms of source-level segregation efficiency and online monitoring, the scale-up limitations and the lack of efficient post-processing course are shortcomings of the system.

Sequeira et al. [12] designed a responsible waste disposal system, SmartBin+, with digital monetization. Their research focused a lot on gamification and user engagement; however, there was not much in the way of integration with routing optimization or waste segregation features.

SentiTrash is an IoT and machine learning-based system that was proposed by Srividya et al. [13] for both urban and rural waste management scenarios. The synergy of sensors and machine learning algorithms was used by the system to categorize and handle waste streams efficiently. Although difficult, the study did not focus on deployment issues at large in metropolitan settings.

Bhavya et al. [14] suggested an IoT-enabled municipal solid waste management system by introducing connected bins that will improve residential waste collection by

curbside waste collection. However, there were no incentive systems integrated into the system and no integration at a macro level.

architecture appears to have succeeded in encouraging user participation with rewards, it has not introduced the issues of waste segregation and route optimization.

Abraham et al. [15] have come up with an RFID-based smart waste disposal system with user incentive mechanisms. Their model automated keeping track of user involvement and hence minimized the need of manual supervision. While the

Paper No.	Dataset used	Objective	Methodology used	Results	Limitations
1	Prototype smart-bin deployment / field sensor logs	Introduce IoT-enabled smart bins for real-time monitoring and alerts	Ultrasonic / IR sensors, microcontroller, cloud reporting	Improved fill-level visibility and collection responsiveness	Focused mainly on monitoring; no incentives or route optimization
2	Campus disposal logs and user interaction records	Promote sustainable campus disposal through rewards (Credibin)	IoT bins + reward allocation, mobile app integration	Increased user participation and responsible disposal on campus	Limited to campus environment; scalability not shown
3	Municipal collection / scheduling records	Optimize waste collection scheduling and decision support	Decision-support + optimization algorithms (scheduling models)	Reduced operational cost and better truck utilization	Algorithmic focus only; lacks segregation and incentive integration
4	Urban collection dataset / service metrics	Ensure fairness in garbage collection across zones	Fairness-oriented resource allocation model using IoT inputs	More equitable service distribution across regions	Did not include real-time routing or citizen engagement features
5	Prototype IoT bin sensor logs (low-cost hardware)	Build an affordable IoT monitoring system for wide deployment	Low-cost sensors + simple gateway + cloud dashboard	Demonstrated feasible, inexpensive monitoring for cities	Basic functionality only (fill-level); no segregation/optimization
6	Bin-level segregation test dataset	Integrate level indicators to support source segregation	Multi-compartment bins with level indicators + IoT telemetry	Improved source segregation and easier downstream processing	No routing/incentive mechanisms; limited system integration
7	Routing traces + IoT fill-level inputs	Improve routing efficiency using IoT data	Routing optimization algorithms fed by real-time bin status	Fuel/time savings and faster collection cycles	Emphasis on routing; lacks community incentives or segregation
8	Sensor dataset for detection/classification	Automate garbage detection and segregation at source	Sensor fusion for detection; classification logic	Accurate detection and automated segregation in prototype	Not integrated with municipal dashboards or large-scale ops
9	Literature / survey datasets and secondary sources	Review IoT applications for sustainable waste management	Systematic survey and conceptual framework synthesis	Broad view of IoT potential and research directions	Conceptual only; no novel deployed system or experiments
10	Smart-dustbin prototype logs	Demonstrate IoT integration for city-scale dustbins	Smart dustbin prototype (sensors, comms, dashboard)	Enabled fill-level monitoring at city sites	Basic feature set; lacks optimization, rewards, and segregation
11	Trash segregation and management sensor logs	Classify and segregate waste using multi-sensor IoT system	Multiple sensors integrated with cloud and SMS alerts.	Real-time segregation status with SMS notifications	Limited to small-scale environment, lacks scalability
12	SmartBin+ pilot data (object detection + app)	Gamify and reward responsible disposal; add vision-based checks	Raspberry Pi, object detection (DL), face/ID, rewards app	Higher engagement and awareness in pilot trials	Scalability, sensor accuracy, privacy/ethics concerns
13	IoT + ML labeled waste dataset (SentiTrash)	Combine IoT and ML for classification in urban & rural settings	ESP32/Arduino sensors + ML classifiers (image/sensor models)	Good classification accuracy across test settings	Metro-scale deployment and compute/resource constraints not fully addressed
14	Curbside pickup records and telemetry	Enhance residential curbside collection using IoT technologies	GPS tracking with IoT bin telemetry and dashboard interface.	Improved curbside collection efficiency and transparency	No incentives or comprehensive integration
15	RFID disposal tracking logs	Implement RFID-based tracking and reward system for disposal	RFID-based tracking integrated with cloud and sensors.	Automated tracking and incentives for disposal actions	Focuses solely on behavior tracking without segregation or routing

III. IDENTIFIED RESEARCH GAPS

Despite the great progress of IoT-enabled waste management, there are some important challenges left, which do not enable the effective use of existing systems in a real world context at a large scale. An examination of the existing literature suggests a number of host of research gaps.

One of the major deficiencies is associated with integration of the features in a system. The majority of studies tend to deal with monitoring, segregation, routing or incentive mechanisms in isolation. For instance, systems under optimization of routings tend not to put enough effort on reward schemes or segregation functionalities while reward-based approaches rarely consider the difficulties of massive logistics of operations. This compartmentalization presents a case for the need for integrative platforms where all important actors within an ecosystem are put together.

Flexibility and scalability are yet another monster problem. Many of the solutions proposed have only been tried in small settings - the campuses of universities, small communities or as pilot prototypes. An in-depth research has not been carried out to extend these systems to the large metropolitan areas with heterogeneous populations, heterogeneous infrastructures and heterogeneous waste streams.

Moreover, there is a huge gap in the community-based, user-centred design methodologies. However, incentives mechanisms come with many variants and are too specific, either for rfid-based systems or point accumulation systems. Few studies have been investigating the long-term user behavior, social acceptability or testing of gamification strategies that would be able to sustain user engagement at scale.

Lastly, data integration and security problems are not taken care of properly. Numerous studies are based on localized sensor-networks with little cloud-connectivity and do not solve interoperability issues between different municipalities or privacy issues around user-associated incentive-data. The development of large-scale, safe, and interoperable data ecosystems is prominent in the available research.

IV. STATE OF THE ART METHODOLOGY

The recent changes in waste management shifted their focus not to the old and fixed collection systems but to new models which are technology-laden and sensor-driven, which imply the Internet of Things (IoT)-based and data-centric approaches. The core of modern systems is smart bins with ultrasonic sensors, infrared sensors or proximity sensors, constantly checking the level of filling, the type of wastes, and sending it to cloud servers in real time. With this paradigm shift, the municipal authorities can substitute the fixed and rather inefficient collection schedules with demand based approaches and as such operational efficiency can be dramatically increased and environmental impact reduced.

One major achievement in this area is the adoption of source segregation of waste. The effectiveness of multi-compartment bins fitted with level sensors and sensor-based sorting technologies in sorting waste into different categories like wet, dry, metallic and electronic has been proved by empirical results. Some of the systems use sensor fusion

methods which combine weight, infrared, and gas sensors, and some use the rule-based algorithms in waste classification. These methods reduce the processing workload of downsizing facilities and increase the recycling rate by enabling a better segregation at the point of disposal.

No sensor-based monitoring complements recent studies that have emphasized on the role of optimization and routing developments. Decision-support models also use the real-time data provided by the bins as they are IoT-enabled to adjust the waste collection schedules dynamically, increase the efficiency of the vehicle route, and decrease fuel consumption. Clustering algorithms to group bins and shortest path algorithms to calculate routes have been applied in numerous researchers with satisfactory outcomes of obtaining economical and time-saving waste collection activities.

Besides the efficiency of operations, there is a growing focus on community involvement and incentive systems to the users. RFID-based identification, rewards credit system, and gamified applications, including Credibin and SmartBin+, are examples of how a set of behavioral economics principles are applied in technology. These ways would ensure proper disposal of wastes as each behavior would be connected to tangible rewards thus the increase in participation rates and long-term community engagement.

To sum up, modern technologies of waste management are increasingly geared towards the use of enhanced sensing equipment, dynamic collection logistics and community involvement. Though individual studies have shown impressive results on such topics of monitoring, segregation, routing, and incentive schemes, another research direction of the future is to integrate such parts into efficacious, multiscale, scalable, and easy-to-use structures.

V. FUTURE SCOPE

The future of smart waste management is expected to be dominated by integrated and scalable frameworks with synergistic combinations between monitoring, segregation, optimized logistics, and community participation. Although current research tends to address these components independently of one another, the future of research is looking at consolidating these elements into holistic systems that can operate effectively at the scale of city level.

More focus will be put on source level sorting, enabled by the introduction of multi compartments bins and low cost sensors, combined with dynamic routing algorithms which adapt to data in real time. Additionally, some strategies beyond incentive payments aimed at facilitating community involvement, such as gamification strategies, are also expected to play an active role in ensuring sustained user engagement. Lastly, interconnected interoperability between the various systems of municipalities, and secure connection with cloud systems, will be key to ensure smart waste management remains dynamic, inclusive and poised for future challenges.

VI. CONCLUSION

In summary, the current state of the art shows that there are substantial advances in the realm of smart waste management systems in regards to automation, monitoring, and optimization of collection processes. Nevertheless, there are still critical gaps in areas of user-centricity, community engagement and integration of holistic solutions that include waste sorting, real-time monitoring, and role-based interfaces. In the future, research and development will be specifically aimed at developing such systems that are not only technically sophisticated but will be easy to use for citizens, municipalities and service providers. The proliferation of such integrated solutions can serve to overcome existing gaps, reducing interest conflicts and generating a more efficient functioning, as well as sustainable urban waste management.

REFERENCES

- [1] S. Mehta and A. Rathour, "Innovative Waste Management: The Efficiency of IoT-Enabled Smart Bins," 2024 7th International Conference on Circuit Power and Computing Technologies (ICCPCT), Kollam, India, 2024, pp. 467-471, doi: 10.1109/ICCPCT61902.2024.10672653. keywords: {Waste management; Costs; Urban areas; Systems architecture; Carbon dioxide; Real-time systems; Software; Smart Cities; Waste Management; Internet of Things (IoT); Ultrasonic Sensors; Embedded Devices; Urban Sanitation},
- [2] M. A. Rahman, S. W. Tan, A. Taufiq Asyhari, I. F. Kumiawan, M. J. F. Alenazi and M. Uddin, "IoT-Enabled Intelligent Garbage Management System for Smart City: A Fairness Perspective," in IEEE Access, vol. 12, pp. 82693-82705, 2024, doi: 10.1109/ACCESS.2024.3412098. keywords: {Waste management; Internet of Things; Monitoring; Waste materials; Smart cities; Real-time systems; Costs; Fairness; garbage collection; IoT; LoRa; smart city; waste management},
- [3] J. A. Batoon, S. M. D. Lainez, R. M. de Angel, A. C. Lagman, R. F. Ramos and V. D. Dorongon, "Waste Management Scheduling Using Optimization and Decision Support Algorithms," 2024 IEEE 15th Control and System Graduate Research Colloquium (ICSGRC), SHAH ALAM, Malaysia, 2024, pp. 222-226, doi: 10.1109/ICSGRC62081.2024.10690878. keywords: {Waste management; Schedules; Reviews; Software algorithms; Prototypes; Scheduling; Mobile handsets; Software; Optimization; Software development management; Component; Garbage Collection; Waste Management; Community; GPS Tracking; and Technology},
- [4] M. V. Rajesh, A. Lakshmanarao, J. S. A. Rongala and C. D. Priyanka, "IoT based smart system for garbage detection and segregation," 2023 IEEE Silchar Subsection Conference (SILCON), Silchar, India, 2023, pp. 1-4, doi: 10.1109/SILCON59133.2023.10405323. keywords: {Economics; Waste management; Urban areas; Robot sensing systems; Acoustics; Recycling; Monitoring; Garbage; IoT; Smart Dustbin; Ultrasonic Sensors},
- [5] M. P. J. E. G S and L. R. B. Raj S, "Garbage monitoring system using IoT," 2024 2nd International Conference on Networking and Communications (ICNWC), Chennai, India, 2024, pp. 1-6, doi: 10.1109/ICNWC60771.2024.10537580. keywords: {Waste management; Wireless sensor networks; Waste materials; Data analysis; Wireless networks; Real-time systems; Servers; Industrial IoT device; actual-time systems; systems of sensors; distant observation; solid refuse management; dust bin; solar-powered},
- [6] M. P. Varghese, V. S. Anooja, R. Akhila, M. Krishnakumar and A. Xavier, "IoT-Based Smart Waste Management System with Level Indicators for Effective Garbage Waste Segregation," 2024 Third International Conference on Distributed Computing and Electrical Circuits and Electronics (ICDCECE), Ballari, India, 2024, pp. 1-5, doi: 10.1109/ICDCECE60827.2024.10548589. keywords: {Waste management; Visualization; Urban areas; Recycling; Internet of Things; Sustainable development; Waste disposal; Garbage Segregation; Internet of Things; Level Indicators; Smart Waste Management; Urban areas},
- [7] R. Tandon and A. Bansal, "IoT-enabled Smart Waste Management: Leveraging the Power of IoT for Sustainable Solutions," 2023 3rd International Conference on Innovative Sustainable Computational Technologies (CISCT), Dehradun, India, 2023, pp. 1-6, doi: 10.1109/CISCT57197.2023.10351210. keywords: {Waste management; Wireless sensor networks; Costs; Sociology; Transportation; Electronic waste; Internet of Things; Internet of things (IoT); Waste Management; Smart Cities; Computer vision; Sustainable development},
- [8] K. S. Abishek, D. Deepak, R. D. Babu and S. Keerthana, "An IoT-based Smart Waste Disposal and Credibin Rewards System for Sustainable Campus Waste Management," 2025 International Conference on Electronics and Renewable Systems (ICEARS), Tuticorin, India, 2025, pp. 674-679, doi: 10.1109/ICEARS64219.2025.10941118. keywords: {Schedules; Renewable energy sources; Scalability; Green products; Sensor systems; Real-time systems; Acoustics; Waste disposal; Sustainable development; Intelligent sensors; IoT; Ultrasonic Sensors; Weight Detection; Real-Time Monitoring; Incentive-Based Reward},
- [9] P. Dharan and V. M. Manikandan, "A Smart Waste Management System with Optimized Routing Method," 2024 15th International Conference on Computing Communication and Networking Technologies (ICCCNT), Kamand, India, 2024, pp. 1-6, doi: 10.1109/ICCCNT61001.2024.10725635. keywords: {Waste management; Technological innovation; Costs; Smart cities; Real-time systems; Acoustics; Sensor systems; Fuels; Wireless fidelity; Intelligent sensors; Smart Dustbin; Arduino MKR Wifi 1010; Ultrasonic sensors; Smart Garbage; Waste collection; Route optimization; Internet of things; Machine learning},
- [10] G. V. V. S. L. Rupesh, M. S. Advytha, N. B. Prakash, V. Sahu, A. Gupta and S. K. Dargar, "A Smart-Dustbin and Integration of Waste Management System using IoT," 2022 4th International Conference on Inventive Research in Computing Applications (ICIRCA), Coimbatore, India, 2022, pp. 393-397, doi: 10.1109/ICIRCA54612.2022.9985021. keywords: {Waste management; Smart cities; Organizations; Filling; Internet of Things; Monitoring; Meteorology; IoT; Embedded; Smart city; Node MCU; Arduino},
- [11] O. Chavan, V. Jaware, D. Doiphode and P. Deshmukh, "IoT-Powered Trash Segregation and Waste Management: An Ingenious Approach to a Sustainable Environment," 2024 IEEE International Conference on Computing, Power and Communication Technologies (IC2PCT), Greater Noida, India, 2024, pp. 595-598, doi: 10.1109/IC2PCT60090.2024.10486767. keywords: {Waste management; Data analysis; Databases; Green products; Real-time systems; Communications technology; Recycling; IoT device; metal waste; wet waste; dry waste segregation; waste management; trash segregation system; etc},
- [12] S. Sequeira, C. Chaugule, A. Castelino, J. DCruz, K. Wagle and A. Pansare, "SmartBin+: Elevating Responsible Waste Disposal with Rewards," 2024 4th International Conference on Sustainable Expert Systems (ICES), Kaski, Nepal, 2024, pp. 296-304, doi: 10.1109/ICES63445.2024.10763196. keywords: {Deep learning; Ethics; Accuracy; Scalability; Face recognition; Urban areas; Object detection; Waste disposal; Expert systems; Monitoring; Raspberry Pi; Object detection; deep learning; reward system; face recognition; environmental awareness},
- [13] K. Srividya, J. Nivedhana, R. Amrutha and M. Vijayaraj, "Effective Sentitrash System for Urban and Rural Environments Using IoT and ML Technology- : A comprehensive solution integrating multiple sensors with ESP32 and Arduino platforms," 2024 International Conference on Power, Energy, Control and Transmission Systems (ICPECTS), Chennai, India, 2024, pp. 1-6, doi: 10.1109/ICPECTS62210.2024.10780305. keywords: {Waste reduction; Visualization; Rain; Costs; Urban areas; Control systems; Real-time systems; Sensor systems; Internet of Things; Intelligent sensors; IoT; Node MCU; Arduino UNO; Multiple sensors (Ultrasonic; rain; gas); buzzer; LED; Bluetooth/Wi-Fi; Blynk App; MC(esp32); Software(Arduino 1.8.2); Arduino cable; bin},
- [14] B. R. B. Rajalakshmi, P. V. R. D. N and G. V. Srushil, "Curbside Waste Management System," 2024 International Conference on Computational Intelligence for Security, Communication and Sustainable Development (CISCSD), Port Blair, India, 2024, pp. 266-269, doi: 10.1109/CISCSD63381.2024.00066. keywords: {Waste management; Waste materials; Technological innovation; Urban areas; Internet of Things; Sustainable development; Monitoring; Smart

phones;Global Positioning System; Vehicles; waste management;GPS;RFID;internet of things;smart bins},

- [15] M. Abraham, S. Sreeraj, D. Milton, A. A. Yohannan and G. J. L. Paulraj, "Reward based Smart Waste Disposal System using RFID," 2023 International Conference on Sustainable Computing and Data Communication Systems (ICSCDS), Erode, India, 2023, pp. 1081-

1084, doi: 10.1109/ICSCDS56580.2023.10104738. keywords: {Deep learning;Pollution;Urban areas;Prototypes;Mobile applications;Data communication;Waste disposal; Mobile application development; Radio Frequency Identification; Internet of Things; Smart garbage collection;Waste Management;Reward based system},

Research Paper Reviewed

contact. wrfer <contact.wrfer@gmail.com>
To: Harshitha Nk <nkharshitha19@gmail.com>

Wed, 26 Nov at 11:13 AM

Dear Author,
Greetings and best wishes for the day !!

Our International Conference has accepted your Research Paper entitled "**A Survey on Smart Waste Management Systems: Trends, Challenges, and Future Directions**" with Paper ID **WR-CSIT-ERNK-261125-13871**

International Conference on Recent Advances in Computer Science and Information Technology

Note: Today is the last day of Registration
There is no additional Fee for Certificates, Proceeding and Publication.

Registration Link: <http://wrfer.org/PAYMENT/>

After the Registration you will get the Acceptance letter and Conference Schedule.

Categories	Non Indian Nationals	Indian Nationals
Authors (WRFER Member with valid Membership ID)	250 USD	4000 INR
Authors Student (Bachelors/Masters)(Non Member)	280 USD	4500 INR
Authors :PhD/Post Doc/Academician/Professional (Non Member)	300 USD	5000 INR
Listeners (Do not have any paper)	100 USD	2000 INR
Additional Page (8 pages allowed with normal registration)	10 USD	500 INR
Extra Proceeding	30 USD	500 INR

[Quoted text hidden]



GLOBAL ACADEMY OF TECHNOLOGY



Department of Computer Science and Engineering

CERTIFICATE OF APPRECIATION

This Certifies that

Mr/Ms Harshitha N K bearing USN 1GA22CS071

has participated in the 'Project-Expo-25' Organised by the
Department of CSE, in Association with ACM, GAT STUDENT
CHAPTER Conducted on 3rd December 2025

SKSwamy

Dr. Kumaraswamy S
Prof and Head
Dept of CSE, GAT



[Signature]

Dr. Balakrishna H.B
Principal, GAT